



High Energy Physics - Theory

Continuous spin superparticle in $4D$, $\mathcal{N} = 1$ curved superspaceI. L. Buchbinder^{a,b,c}, S. A. Fedoruk^{id a,*}^a Bogoliubov Laboratory of Theoretical Physics, Joint Institute for Nuclear Research, 141980, Dubna, Moscow, Russia^b Center of Theoretical Physics, Tomsk State Pedagogical University, 634041, Tomsk, Russia^c National Research Tomsk State University, 634050, Tomsk, Russia

ARTICLE INFO

PACS:

11.10.Ef

11.30.-j

11.30.Cp

03.65.Pm

02.40.Ky

Keywords:

Continuous spin particles

Anti-de Sitter superspaces

Constrained supersymmetric mechanics

ABSTRACT

We present a new particle model that describes the dynamics of a $4D$, $\mathcal{N}=1$ continuous spin particle in AdS_4 superspace and is a generalization of the continuous-spin superparticle model in flat $4D$, $\mathcal{N}=1$ superspace proposed in 2506.19709 [hep-th]. The model is described by $4D$, $\mathcal{N}=1$ superspace coordinates together with commuting spinor additional variables, which are inherent ingredients of continuous spin models. The Lagrangian and the system of four bosonic and four fermionic phase space constraints are derived. The consistency condition for constraints imposes a restriction on supergeometry to be AdS superspace. It is shown that the bosonic constraints are first-class constraints. A covariant procedure based on the use of additional variables is developed to divide the four fermionic constraints into first and second classes. It is proved that, unlike the flat case, only one fermionic constraint is a first-class constraint, while the other three are second-class constraints. In the flat limit, one of these second-class constraints becomes a first-class one.

1. Introduction

One-dimensional models describing the dynamics of relativistic particles in background spaces, in addition to being interesting in themselves due to many remarkable properties, are a testing ground for constructing more complicated gauge field theories and for investigating background geometry. These models have a variety of global and local (super)symmetries and can serve as a basis for developing new promising multidimensional (super)gauge theories. At present, the investigation of relativistic particle models has developed into a fairly vast trend of research in theoretical and mathematical physics. The number of publications on this topic apparently amounts to several hundred.

Among the various models of relativistic particles, one can single out the superparticle models whose world lines lie in flat or curved superspaces. Such models are widely discussed in the literature, and many of their classical and quantum aspects have been studied in detail. Within the extensive literature on superparticle models, we will only mention those publications that are somehow relevant to our present paper [1–20]. Later, we will refer to these papers in more detail at the appropriate points of our work. Note, however, that all these papers study only superparticle models with definite superspins or superhelicities. The bosonic sector of the above models in flat space corresponds to irreducible finite spin (or helicity) representations of the Poincaré group.

Among a number of papers on superparticles we especially mention the papers [5,8–10,16] (see also the review [13]) where superparticle models were studied in the curved superspace. Let us also note the papers [14,15,19,20], which considered the superparticle in the AdS superspace. The models under consideration have a remarkable gauge structure that in the canonical approach manifests itself in the appearance of first-class bosonic constraints and specific fermion constraints with a not evident and non-trivial

* Corresponding author.

E-mail addresses: buchbinder@theor.jinr.ru (I.L. Buchbinder), fedoruk@theor.jinr.ru (S.A. Fedoruk).

division into first and second classes. Note that for Lorentz-covariant division of such constraints it is necessary to introduce additional bosonic variables, which are often absent in the original Lagrangian. It is worth pointing out that the consistency condition for first-class constraints imposes a restriction on the geometry of curved superspace in the form of supergravity equations. It was also found that a similar result occurs in superstring theory in the supergravity background (see, e.g., the review [13]).

Until recently, the continuous spin representations of the Poincaré group have not been studied in the context of superparticle models. Recently in [21] we constructed a novel relativistic particle model that for the first time describes the propagation of a continuous spin superparticle in $4D$, $\mathcal{N}=1$ flat superspace. An approach was based on a supersymmetric generalization of our earlier works where the bosonic continuous spin particle models were constructed in flat and curved space-times [22–24].¹ A remarkable feature of continuous spin models is that in such models additional bosonic variables are an obligatory part of the coordinates. Using the commuting spinors as these additional coordinates introduced in works [22–24] allowed us, in the superparticle model [21], to divide in a natural way the fermion constraints in phase space into first- and second-class constraints. In the present paper, we propose a generalization of the model [21] and formulate a new Lagrangian continuous spin superparticle model in the curved $4D$, $\mathcal{N}=1$ superspace corresponding to the supergravity background (see, e.g., [42,43]). As we will see, an additional bosonic spinor coordinates play an extremely non-trivial role in the division of fermionic constraints into first and second class and, besides, the model turns out to be consistent only in AdS superspace. The aspects of AdS symmetry needed for our purposes were considered in [44], where AdS superspace was first presented in explicit form.²

The paper is organized as follows. In Section 2 we briefly describe the continuous spin model in flat $4D$, $\mathcal{N}=1$ superspace derived earlier in [21]. Section 3 is devoted to problems of generalizing the results of [21] to $4D$, $\mathcal{N}=1$ curved superspace using the technique of differential supergeometry [43]. It is pointed out that a direct generalization turns out to be too naive and leads to different inconsistencies and requires a modification. In Section 4 we first show that the theory under consideration can consistently be formulated only in AdS superspace and second we derive consistent bosonic and fermionic constraints and a consistent Lagrangian. It is shown that the modified bosonic constraints form the first-class constraints. As to fermionic constraints, their class is not evident from the very beginning. We have developed a procedure for dividing the fermionic constraints into first- and second-class constraints using additional bosonic spinor variables that are inherent in continuous spin models. It is shown that four fermionic constraints are divided into three second-class constraints and one first-class constraint. This is significantly different from the superparticle model in flat superspace, where four fermion constraints are divided into two second-class constraints and two first-class constraints. In the flat limit, one of the fermionic second-class constraints transforms into a first-class constraint. In Section 5, using the first-class fermionic constraint as a generator of gauge transformations in phase space, we derive κ -symmetry transformations for the continuous spin superparticle model in curved superspace. Note that the auxiliary variables are also transformed under κ -symmetry transformations in the case under consideration. The Summary is devoted to the discussion of the results obtained. Some auxiliary technical details are given in the Appendices.

2. Continuous spin superparticle in flat superspace

Before constructing the superparticle model in curved superspace, we describe the infinite spin particle in $4D$ flat target superspace. To do this, we use the anticommuting Weyl spinor fields θ^α , $\bar{\theta}_{\dot{\alpha}}$, which, together with the bosonic coordinates x^a , form the $4D$, $\mathcal{N}=1$ flat superspace with the coordinates $z^A = (x^a, \theta^\alpha, \bar{\theta}_{\dot{\alpha}})$. In describing the continuous spin particle we use additional spinor coordinates ξ^α , $\bar{\xi}^{\dot{\alpha}} = (\xi^\alpha)^*$ and π_α , $\bar{\pi}_{\dot{\alpha}} = (\pi_\alpha)^*$. The latter of them play the role of momenta for the former.

In [21], we derived the target space supersymmetric-invariant Lagrangian in the form

$$L_0 = \frac{1}{2e} W^a W_a + \nabla_{\xi}^{\xi^\alpha} \pi_\alpha + \bar{\pi}_{\dot{\alpha}} \nabla_{\bar{\xi}}^{\bar{\xi}^{\dot{\alpha}}} - (\lambda + \bar{\lambda}) \mu. \quad (2.1)$$

Here the four-vector

$$W^a := \omega^a + \lambda(\xi \sigma^a \bar{\xi}) + \bar{\lambda}(\bar{\pi} \bar{\sigma}^a \pi) \quad (2.2)$$

is a continuous spin generalization of the Cartan ω -superforms

$$\omega^a = \dot{x}^a + i\theta \sigma^a \dot{\bar{\theta}} - i\bar{\theta} \sigma^a \dot{\theta}, \quad (2.3)$$

which are invariant with respect to supersymmetry transformations

$$\delta x^a = i\theta \sigma^a \bar{\epsilon} - i\bar{\theta} \sigma^a \epsilon, \quad \delta \theta^\alpha = \epsilon^\alpha, \quad \delta \bar{\theta}_{\dot{\alpha}} = \bar{\epsilon}_{\dot{\alpha}} \quad (2.4)$$

with the anticommuting global parameters ϵ^α , $\bar{\epsilon}^{\dot{\alpha}} = (\epsilon^\alpha)^*$. Note that the bosonic auxiliary coordinates and momenta are inert with respect to supersymmetry transformations: $\delta \xi^\alpha = 0$, $\delta \bar{\xi}^{\dot{\alpha}} = 0$, $\delta \pi_\alpha = 0$, $\delta \bar{\pi}_{\dot{\alpha}} = 0$.

In (2.1), x^a , ξ^α , $\bar{\xi}^{\dot{\alpha}}$, π_α , $\bar{\pi}_{\dot{\alpha}}$ are $1D$ fields, where τ is the evolution parameter: $x^a = x^a(\tau)$, $\xi^\alpha = \xi^\alpha(\tau)$, etc. We use the standard notation for τ -derivatives: $\dot{x}^a$, $\dot{\xi}^\alpha$, $\dot{\bar{\xi}}^{\dot{\alpha}}$. The parameter $\mu \in \mathbb{R}$ in (2.1) is a nonzero constant and $e(\tau)$, $\lambda(\tau)$, $\bar{\lambda}(\tau)$, $A(\tau)$ are the Lagrange multipliers.

¹ Relatively recently, a direction has emerged in field theory associated with the field description of irreducible massless representations of the Poincaré group with continuous spin. Among a large number of works in this direction, we note only some papers of the last few years (see, e.g., [25–35] and references therein). The papers [21–24] can be considered as a natural part of this direction. Supersymmetric continuous spin field models were discussed, e.g., in the recent papers [36–41].

² For other pioneering studies on AdS supersymmetry see in [45–48].

In addition, the quantities

$$\nabla \xi^\alpha = \dot{\xi}^\alpha + i A \xi^\alpha, \quad \nabla \bar{\xi}^{\dot{\alpha}} = \dot{\bar{\xi}}^{\dot{\alpha}} - i A \bar{\xi}^{\dot{\alpha}} \quad (2.5)$$

are the U(1)-covariant derivatives.

The model with the Lagrangian (2.1) is described by the fermionic constraints

$$D_\alpha = i p_\alpha - (\sigma^a \bar{\theta})_\alpha p_a \approx 0, \quad \bar{D}_{\dot{\alpha}} = -(D_\alpha)^* = i \bar{p}_{\dot{\alpha}} + (\theta \sigma^a)_{\dot{\alpha}} p_a \approx 0, \quad (2.6)$$

and the bosonic constraints

$$l_0 = p_a p^a \approx 0, \quad (2.7)$$

$$l = p_a (\xi \sigma^a \bar{\xi}) - \mu \approx 0, \quad (2.8)$$

$$\tilde{l} = p_a (\bar{\pi} \tilde{\sigma}^a \pi) - \mu \approx 0, \quad (2.9)$$

$$u = N - \bar{N} \approx 0, \quad (2.10)$$

where

$$N = \xi^\alpha \pi_\alpha, \quad \bar{N} = \bar{\pi}_{\dot{\alpha}} \bar{\xi}^{\dot{\alpha}}. \quad (2.11)$$

In (2.6)–(2.10), the quantities p_a , p_a , $\bar{p}^{\dot{\alpha}}$ are momenta for x^a , θ^α , $\bar{\theta}_{\dot{\alpha}}$ respectively. The canonical commutational relations are defined in terms of nonzero Poisson brackets of the canonical variables:

$$\{p_A, z^B\} = -\delta_A^B, \quad \{x^a, p_b\} = \delta_b^a, \quad \{\theta^\alpha, p_\beta\} = -\delta_\beta^\alpha, \quad \{\bar{\theta}_{\dot{\alpha}}, \bar{p}^{\dot{\beta}}\} = -\delta_{\dot{\alpha}}^{\dot{\beta}}. \quad (2.12)$$

$$\{\xi^\alpha, \pi_\beta\} = \delta_\beta^\alpha, \quad \{\bar{\xi}^{\dot{\alpha}}, \bar{\pi}_{\dot{\beta}}\} = \delta_{\dot{\beta}}^{\dot{\alpha}}. \quad (2.13)$$

Using (2.12) and (2.12), we obtain that the non-vanishing Poisson brackets of fermionic constraints (2.6) and bosonic constraints (2.7)–(2.10) have the form

$$\{D_\alpha, \bar{D}_{\dot{\alpha}}\} = -2i \sigma_{\alpha\dot{\alpha}}^a p_a, \quad (2.14)$$

$$\{l, \tilde{l}\} = -K l_0, \quad (2.15)$$

where

$$K := N + \bar{N} \quad (2.16)$$

and N and $\bar{N}$ are given by (2.11).

Using the expressions (2.14) and (2.15), we find that all bosonic constraints (2.7)–(2.10) are the first-class constraints, whereas four fermionic constraints (2.6) are the mixture of two first-class constraints and two second-class constraints. The first-class constraints are constructed on the basis of fermionic constraints by convolving them with the matrix $\bar{p}^{\dot{\alpha}\alpha}$. That is, the constraints $\bar{p}^{\dot{\alpha}\alpha} D_\alpha \approx 0$, $\bar{D}_{\dot{\alpha}} \bar{p}^{\dot{\alpha}\alpha} \approx 0$ are the first-class ones which are the generators of local fermionic κ -invariance. But these constraints are not independent due to the constraint (2.7). It is impossible to extract independent first- and second-class constraints from the constraints (2.6) in a covariant way without using additional commuting variables.

However, a remarkable feature of the model under consideration is that it already contains the commuting spinor variables ξ^α , $\bar{\xi}^{\dot{\alpha}}$ by construction. This opens the opportunity of constructing independent fermionic first- and second-class constraints in a covariant way using these variables (see, e.g., a similar consideration in [17]). As a result, the odd constraints (2.6) split into two second-class constraints and two first-class constraints as follows. The fermionic second-class constraints look like

$$G := \xi^\alpha D_\alpha \approx 0, \quad \bar{G} := \bar{D}_{\dot{\alpha}} \bar{\xi}^{\dot{\alpha}} \approx 0, \quad \bar{G} = -(G)^* \quad (2.17)$$

while the fermionic first-class constraints are

$$F := \bar{\xi}_{\dot{\alpha}} \bar{p}^{\dot{\alpha}\alpha} D_\alpha \approx 0, \quad \bar{F} := \bar{D}_{\dot{\alpha}} \bar{p}^{\dot{\alpha}\alpha} \xi_\alpha \approx 0, \quad \bar{F} = -(F)^* \quad (2.18)$$

The algebra of constraints (2.17) and (2.18) has the form

$$\{G, \bar{G}\} = -2i l - 2i \mu, \quad \{F, \bar{F}\} = 2i (\xi p \bar{\xi}) l_0, \quad (2.19)$$

$$\{G, \tilde{l}\} = F, \quad \{\bar{G}, \tilde{l}\} = \bar{F}, \quad \{F, \tilde{l}\} = \pi^\alpha D_\alpha l_0, \quad \{\bar{F}, \tilde{l}\} = \bar{D}_{\dot{\alpha}} \bar{\pi}^{\dot{\alpha}} l_0, \quad (2.20)$$

$$\{G, u\} = G, \quad \{\bar{G}, u\} = -\bar{G}, \quad \{F, u\} = -F, \quad \{\bar{F}, u\} = \bar{F}. \quad (2.21)$$

It is clearly seen that the only second-class constraints are G and $\bar{G}$.

Let us describe the main points of the procedure for extracting first- and second-class constraints from fermionic constraints (2.6). First, due to the constraints (2.7) and (2.8), i. e. $p^2 \approx 0$ and $\xi p \bar{\xi} \approx \mu \neq 0$, we have the identities

$$\delta_\alpha^\beta = \frac{1}{\mu} [\xi_\alpha (\bar{\xi} \bar{p})^\beta + (p \bar{\xi})_\alpha \xi^\beta], \quad \delta_{\dot{\alpha}}^{\dot{\beta}} = \frac{1}{\mu} [\bar{\xi}_{\dot{\alpha}} (\bar{p} \bar{\xi})^{\dot{\beta}} + (\xi p)_{\dot{\alpha}} \bar{\xi}^{\dot{\beta}}]. \quad (2.22)$$

Using these identities, we present the constraints (2.6) in the form

$$D_\alpha \approx \frac{1}{\mu} [(p \bar{\xi})_\alpha G + \xi_\alpha F], \quad \bar{D}_{\dot{\alpha}} \approx \frac{1}{\mu} [(\xi p)_{\dot{\alpha}} \bar{G} + \bar{\xi}_{\dot{\alpha}} \bar{F}].$$

We see that the constraints G and F are equivalent to D_α and the constraints $\bar{G}$ and $\bar{F}$ are equivalent to $\bar{D}_\alpha$.

The canonical Hamiltonian in the considered system is defined as a linear combination of all first-class constraints:

$$H_0 = \frac{1}{2} e l_0 - \lambda l - \tilde{\lambda} \tilde{l} - i A u + \chi F + \bar{\chi} \bar{F}. \quad (2.23)$$

where the constraints e , λ , $\tilde{\lambda}$, u , χ and $\bar{\chi}$ are the Lagrange multipliers.

3. The continuous spin superparticle in curved superspace: problem of Lagrangian and constraints

We begin to consider the superparticle in a curved superspace parameterized by supercoordinates

$$z^M = (x^m, \theta^\mu, \bar{\theta}_{\dot{\mu}}), \quad \text{where} \quad x^m = (x^m)^*, \quad \bar{\theta}^{\dot{\mu}} = (\theta^\mu)^*, \quad \bar{\theta}^{\dot{\mu}} = \epsilon^{\dot{\mu}\nu} \bar{\theta}_{\dot{\nu}}. \quad (3.1)$$

Also, the infinite spin particle is characterized by additional target space coordinates which are the commuting Weyl spinors ξ^α , $\bar{\xi}^{\dot{\alpha}}$, π_α , $\bar{\pi}_{\dot{\alpha}}$ with flat indices like in the previous section.

As a first step, we perform a direct curved superspace generalization of the Lagrangian (2.1). For this, we make the following replacements in (2.1):

$$\omega^a \rightarrow z^M E_M^a, \quad \xi^\alpha \rightarrow \xi^\alpha + z^M \Omega_M^{\alpha\beta} \xi_\beta, \quad \bar{\xi}^{\dot{\alpha}} \rightarrow \bar{\xi}^{\dot{\alpha}} + z^M \Omega_M^{\dot{\alpha}\dot{\beta}} \bar{\xi}_{\dot{\beta}}, \quad (3.2)$$

where $E_M^A(z)$ is the supervielbein matrix and $\Omega_M^{\alpha\beta}(z)$, $\Omega_M^{\dot{\alpha}\dot{\beta}}(z)$ are the components of spin superconnection (see, e.g., [43]). Let us turn attention to the rules of contraction in the superindex M in (3.2): $A^M B_M = A^m B_m + A^\mu B_\mu + A_{\dot{\mu}} B^{\dot{\mu}}$. For example,

$$z^M E_M^a = x^m E_m^a + \theta^\mu E_\mu^a + \bar{\theta}_{\dot{\mu}} E^{\dot{\mu}a}.$$

Thus, at this stage the curved superspace generalization of the supersymmetric-invariant Lagrangian (2.1) has the form

$$L = \frac{1}{2e} \mathcal{W}^a \mathcal{W}_a + D\xi^\alpha \pi_\alpha + \bar{\pi}_{\dot{\alpha}} D\bar{\xi}^{\dot{\alpha}} - (\lambda + \tilde{\lambda}) \mu, \quad (3.3)$$

where now (see (2.2) for comparison)

$$\mathcal{W}^a := z^M E_M^a + \lambda(\xi^\alpha \bar{\xi}_\alpha) + \tilde{\lambda}(\bar{\pi}_{\dot{\alpha}} \pi^{\dot{\alpha}}) \quad (3.4)$$

and (see (2.5) for comparison)

$$D\xi^\alpha := \dot{\xi}^\alpha + z^M \Omega_M^{\alpha\beta} \xi_\beta + i A \xi^\alpha, \quad D\bar{\xi}^{\dot{\alpha}} := \dot{\bar{\xi}}^{\dot{\alpha}} + z^M \Omega_M^{\dot{\alpha}\dot{\beta}} \bar{\xi}_{\dot{\beta}} - i A \bar{\xi}^{\dot{\alpha}}. \quad (3.5)$$

In the flat case, when the vielbein $E_M^A(z)$ has only the following nonzero components³

$$\overset{\circ}{E}_M^A = \left(\overset{\circ}{E}_m^a = \delta_m^a, \overset{\circ}{E}_\mu^a = \delta_\mu^a, \overset{\circ}{E}_{\dot{\alpha}}^{\dot{\mu}} = \delta_{\dot{\alpha}}^{\dot{\mu}}, \overset{\circ}{E}_\mu^{\dot{\mu}} = -i(\sigma^a)_{\mu\dot{\mu}} \bar{\theta}^{\dot{\mu}}, \overset{\circ}{E}^{\dot{\mu}a} = -i(\bar{\sigma}^a)^{\dot{\mu}\mu} \theta_\mu \right)$$

and the spin connection is zero, the Lagrangian (3.3) becomes the flat space Lagrangian (2.1).

To construct a canonical formulation, one introduces the supermomenta p_M corresponding to the supercoordinates z^M :

$$p_M = \frac{\partial L}{\partial z^M} : \quad p_m = \frac{\partial L}{\partial x^m}, \quad p_\mu = \frac{\partial L}{\partial \theta^\mu}, \quad \bar{p}^{\dot{\mu}} = \frac{\partial L}{\partial \bar{\theta}_{\dot{\mu}}}. \quad (3.6)$$

Relations (3.6) yield

$$p_M = \frac{1}{e} E_M^a \mathcal{W}_a + \Omega_M^{\alpha\beta} \pi_\alpha \xi_\beta + \Omega_M^{\dot{\alpha}\dot{\beta}} \bar{\pi}_{\dot{\alpha}} \bar{\xi}_{\dot{\beta}}. \quad (3.7)$$

The canonical Poisson brackets of the supermomenta p_M and the supercoordinates z^M look like (see (2.12))⁴

$$\{p_M, z^N\} = -\delta_M^N : \quad \{p_m, x^n\} = -\delta_m^n, \quad \{p_\mu, \theta^\nu\} = -\delta_\mu^\nu, \quad \{\bar{p}^{\dot{\mu}}, \bar{\theta}_{\dot{\nu}}\} = -\delta_{\dot{\mu}}^{\dot{\nu}}. \quad (3.8)$$

The commuting Weyl spinors ξ^α , $\bar{\xi}^{\dot{\alpha}}$ and π_α , $\bar{\pi}_{\dot{\alpha}}$ form canonical pairs: they nonvanishing Poisson brackets have the form (2.13), i.e.

$$\{\pi_\beta, \xi^\alpha\}_D = -\delta_\beta^\alpha, \quad \{\bar{\pi}_{\dot{\beta}}, \bar{\xi}^{\dot{\alpha}}\}_D = -\delta_{\dot{\beta}}^{\dot{\alpha}}. \quad (3.9)$$

With respect to complex conjugation, the momenta transform as follows:

$$(p_m)^* = p_m, \quad (p_\mu)^* = \bar{p}_{\dot{\mu}} = \epsilon_{\dot{\mu}\nu} \bar{p}^{\dot{\nu}}, \quad (\pi_\alpha)^* = \bar{\pi}_{\dot{\alpha}}. \quad (3.10)$$

It is convenient to represent expressions (3.7) in the form:

$$p_M = \frac{1}{e} E_M^a \mathcal{W}_a, \quad (3.11)$$

where

$$p_M := p_M - \Omega_M^{\alpha\beta} \pi_\alpha \xi_\beta - \Omega_M^{\dot{\alpha}\dot{\beta}} \bar{\pi}_{\dot{\alpha}} \bar{\xi}_{\dot{\beta}}. \quad (3.12)$$

³ Here $\theta_\mu = \epsilon_{\mu\nu} \theta^\nu$ and $(\bar{\sigma}^a)^{\dot{\mu}\mu} = \epsilon^{\dot{\mu}\nu} \epsilon^{\mu\psi} (\sigma^a)_{\nu\psi}$

⁴ For convenience we use the notation for the momentum corresponding to $\bar{\theta}_{\dot{\mu}}$ with a bar, that is $\bar{p}^{\dot{\mu}}$.

We rewrite the $\mathcal{P}_M$ as follows:

$$\mathcal{P}_M = p_M + \frac{1}{2} \Omega_M^{ab} M_{ab}, \quad (3.13)$$

where

$$M_{ab} = \xi \sigma_{ab} \pi - \bar{\pi} \tilde{\sigma}_{ab} \bar{\xi}, \quad M_{ab} = (M_{ab})^* \quad (3.14)$$

are the angular momenta generators forming the Lorentz algebra

$$\{M_{ab}, M_{cd}\} = \eta_{ac} M_{bd} + \eta_{bd} M_{ac} - \eta_{ad} M_{bc} - \eta_{bc} M_{ad}. \quad (3.15)$$

The superconnections with vector and spinor indices relate to each other by:

$$\Omega_M^{ab} = (\sigma^{ab})_{\alpha\beta} \Omega_M^{\alpha\beta} - (\tilde{\sigma}^{ab})_{\dot{\alpha}\dot{\beta}} \Omega_M^{\dot{\alpha}\dot{\beta}}, \quad (3.16)$$

$$\Omega_M^{\alpha\beta} = \frac{1}{2} \Omega_M^{ab} (\sigma_{ab})^{\alpha\beta}, \quad \Omega_M^{\dot{\alpha}\dot{\beta}} = -\frac{1}{2} \Omega_M^{ab} (\tilde{\sigma}_{ab})^{\dot{\alpha}\dot{\beta}}. \quad (3.17)$$

The Poisson brackets of quantities (3.13) look like

$$\{\mathcal{P}_M, \mathcal{P}_N\}_D = -\frac{1}{2} R_{MN}^{ab} M_{ab}, \quad (3.18)$$

where⁵

$$R_{MN}^{ab} = \partial_M \Omega_N^{ab} - (-)^{MN} \partial_N \Omega_M^{ab} + \Omega_M^{ac} \Omega_{Nc}^b - (-)^{MN} \Omega_N^{ac} \Omega_{Mc}^b \quad (3.19)$$

is the superspace curvature tensor.

Introducing the inverse supervielbein

$$E_A^M(z), \quad E_M^A E_A^N = \delta_M^N, \quad E_A^M E_M^B = \delta_A^B,$$

we rewrite the quantities (3.13) with world superindices through the quantities with tangent (flat) superindices:

$$\mathcal{P}_A = E_A^M \mathcal{P}_M = p_A + \frac{1}{2} \Omega_A^{ab} M_{ab}, \quad (3.20)$$

where

$$p_A = E_A^M p_M, \quad \Omega_A^{ab} = E_A^M \Omega_M^{ab}. \quad (3.21)$$

The Poisson brackets of the quantities (3.20) are⁶

$$\{\mathcal{P}_A, \mathcal{P}_B\}_D = -\frac{1}{2} R_{AB}^{cd} M_{cd} - C_{AB}^C \mathcal{P}_C, \quad (3.22)$$

where

$$\begin{aligned} R_{AB}^{cd} &= (-)^{M(N+B)} E_A^M E_B^N R_{MN}^{cd} \\ &= E_A^N \partial_N \Omega_B^{cd} - (-)^{AB} E_B^N \partial_N \Omega_A^{cd} + \Omega_A^{ch} \Omega_{Bh}^d - (-)^{AB} \Omega_B^{ch} \Omega_{Ah}^d - C_{AB}^D \Omega_D^{cd} \end{aligned} \quad (3.23)$$

is the target space curvature tensor and

$$C_{AB}^C = \left(E_A^N \partial_N E_B^M - (-)^{AB} E_B^N \partial_N E_A^M \right) E_M^C \quad (3.24)$$

is the connection-free part of the torsion. Note that

$$C_{AB}^C = (-)^{M(N+B)} E_A^M E_B^N C_{MN}^C, \quad \text{where} \quad C_{MN}^A = - \left(\partial_M E_N^A - (-)^{MN} \partial_N E_M^A \right). \quad (3.25)$$

Besides, below we will use the identity [43]

$$\frac{1}{2} R_{AB}^{cd} M_{cd} = R_{AB}^{\gamma\delta} M_{\gamma\delta} + R_{AB}^{\dot{\gamma}\dot{\delta}} \bar{M}_{\dot{\gamma}\dot{\delta}}, \quad (3.26)$$

where, similarly to (3.16), the spin-tensor components of the angular momentum tensor (3.14) defined by

$$M^{ab} = (\sigma^{ab})_{\alpha\beta} M^{\alpha\beta} - (\tilde{\sigma}^{ab})_{\dot{\alpha}\dot{\beta}} \bar{M}^{\dot{\alpha}\dot{\beta}}, \quad (3.27)$$

are equal to

$$M^{\alpha\beta} = -\xi^{(\alpha} \pi^{\beta)}, \quad \bar{M}^{\dot{\alpha}\dot{\beta}} = -\bar{\xi}^{(\dot{\alpha}} \bar{\pi}^{\dot{\beta})}, \quad \bar{M}^{\dot{\alpha}\dot{\beta}} = (M^{\alpha\beta})^*. \quad (3.28)$$

⁵ Here, as usual, M and N in the exponent are equal to 0 or 1 for vector and spinor indices, respectively.

⁶ Compared to the notation in our paper [23], here the definition of the ‘angolonomy coefficients’ C_{AB}^C and supertorsion components T_{AB}^C contain an additional factor of 2, as is customary in the literature on supersymmetry (see [42,43]). In addition, we use an additional sign “-” in the definition of the ‘angolonomy coefficients’ C_{AB}^C and supertorsion components T_{AB}^C . As result, our definitions coincide with the definitions in [43] (in [42], the definition of supertorsion differs by the sign “-” compared to the definition in [43]).

In terms of the quantities (3.20), expressions (3.11) take the form

$$\mathcal{P}_a = \frac{1}{e} \mathcal{W}_a, \quad (3.29)$$

$$\mathcal{P}_\alpha = 0, \quad \bar{\mathcal{P}}^\alpha = 0. \quad (3.30)$$

Similarly to (3.10), the covariant momenta transform under complex conjugation as follows

$$(\mathcal{P}_a)^* = \mathcal{P}_a, \quad (\mathcal{P}_\alpha)^* = \bar{\mathcal{P}}_\alpha = \epsilon_{\alpha\dot{\beta}} \bar{\mathcal{P}}^{\dot{\beta}}. \quad (3.31)$$

Using the Lagrangian (3.3), we obtain the canonical Hamiltonian in the form (see (2.23) for comparison)

$$H_C = \frac{1}{2} e \ell_0 - \lambda \ell - \bar{\lambda} \bar{\ell} - i A u, \quad (3.32)$$

where (2.7)–(2.10)

$$\ell_0 := \mathcal{P}_a \mathcal{P}^a, \quad (3.33)$$

$$\ell := \mathcal{P}_a (\xi \sigma^a \bar{\xi}) - \mu, \quad (3.34)$$

$$\bar{\ell} := \mathcal{P}_a (\bar{\pi} \bar{\sigma}^a \pi) - \mu, \quad (3.35)$$

$$u := N - \bar{N}. \quad (3.36)$$

In (3.32), the quantities e , λ , $\bar{\lambda}$ and A are the Lagrange multipliers (the procedure for processing them and their momenta is exactly the same as for the flat space superparticle considered in the previous section). It is evident that the quantities (3.33)–(3.36) define the bosonic constraints

$$\ell_0 \approx 0, \quad \ell \approx 0, \quad \bar{\ell} \approx 0, \quad u \approx 0. \quad (3.37)$$

In addition to these bosonic constraints, the definition of spinor momenta gives us fermionic constraints (3.30):

$$\mathcal{D}_\alpha := i \mathcal{P}_\alpha \approx 0, \quad \bar{\mathcal{D}}^{\dot{\alpha}} := i \bar{\mathcal{P}}^{\dot{\alpha}} \approx 0. \quad (3.38)$$

In the flat case, when inverse vielbein $E_A{}^M(z)$ has only the following nonzero components

$$E_A{}^M = \left(\overset{\circ}{E}_a{}^m = \delta_a^m, \overset{\circ}{E}_\alpha{}^\mu = \delta_\alpha^\mu, \overset{\circ}{E}_{\dot{\alpha}}{}^{\dot{\mu}} = \delta_{\dot{\alpha}}^{\dot{\mu}}, \overset{\circ}{E}_\alpha{}^m = i(\sigma^m)_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}}, \overset{\circ}{E}_{\dot{\alpha}}{}^{\dot{\mu}} = i(\bar{\sigma}^{\dot{\mu}})^{\dot{\alpha}\alpha} \theta_\alpha \right), \quad (3.39)$$

spin connection vanishes and $\overset{\circ}{P}_m = p_m$, $\overset{\circ}{P}_\mu = p_\mu$, $\overset{\circ}{P}_{\dot{\mu}} = p_{\dot{\mu}}$. Thus, in the flat limit the equalities $\overset{\circ}{P}_a = p_a$, $\overset{\circ}{P}_\alpha = -i D_\alpha$, $\overset{\circ}{P}_{\dot{\alpha}} = \epsilon_{\dot{\alpha}\beta} \overset{\circ}{P}^{\dot{\beta}} = -i \bar{D}_{\dot{\alpha}}$ take place and the constraints (3.38) become the flat fermionic constraints (2.6).

To sum up the interim results, we have arrived at the Hamiltonian of superparticle in curved superspace (3.32) on the basis of Lagrangian (3.3) and found the system of corresponding bosonic (3.37) and fermionic (3.38) constraints. Since the Lagrangian was obtained purely formally only by supercovariantizing the flat superspace Lagrangian, we cannot state that the Lagrangian (3.3) is a correct Lagrangian for the infinite spin superparticle in curved superspace. The matter is that the formal covariantization does not take into account possible non-minimal terms which can be present in true Lagrangian. Therefore, the constraints obtained on the basis of Lagrangian (3.3) cannot be considered as correct constraints for the superparticle in curved superspace. In particular, it is unclear to which classes the constraints given by relations (3.37) and (3.38) belong in the correct theory. This should be extremely important for a complete description of the canonical structure of the model under consideration. Thus, our next stage is to establish a correct form of the first- and second-class constraints in our model without changing the total number of constraints. Taking into account the results on purely bosonic infinite spin particle [23], one can expect that the bosonic constraints (3.37) should be modified in some way by supergeometry dependent terms to get a system of correct first-class constraints. Besides, one can expect some restrictions on supergeometry. As to fermionic constraints (3.38), they should be decomposed in an explicit supercovariant form into first- and second-class constraints as it was done in the previous section. That is, the following requirements must be fulfilled:

1. The bosonic constraints (3.37) must be modified to get the first-class constraints.
2. The fermionic constraints (3.38) must be decomposed into first- and second-class constraints.
3. No additional constraints in the curved superspace should arise.

The realization of these requirements will be considered in the next sections.⁷

4. Curved background superspace construction of the Lagrangian and constraints

4.1. Construction of the bosonic first-class constraints

In this subsection we will discuss a modification of the bosonic constraints to get a system of first-class constraints. Let us begin with calculating the Poisson brackets of the constraints ℓ (3.34) and $\bar{\ell}$ (3.35). The direct calculations lead to

$$\{\ell, \bar{\ell}\} = -K \ell'_0 - \frac{1}{4} (\xi \sigma^{ab} \pi + \bar{\pi} \bar{\sigma}^{ab} \bar{\xi}) R_{ab}{}^{cd} M_{cd} u - (\xi \sigma^a \bar{\xi}) (\bar{\pi} \bar{\sigma}^b \pi) T_{ab}{}^C P_C, \quad (4.1)$$

⁷ Note that the modifications of the constraints in fact mean a modification of the Lagrangian (3.3).

where we have used the relation⁸

$$(\xi\sigma^{[a}\bar{\xi})(\bar{\pi}\tilde{\sigma}^b\pi) = -\frac{1}{2}K M^{ab} + \frac{1}{2}u(\xi\sigma^{ab}\pi + \bar{\pi}\tilde{\sigma}^{ab}\bar{\xi}), \quad (4.2)$$

and the following notation has been introduced:

$$K := N + \bar{N}, \quad (4.3)$$

$$\ell'_0 := \ell_0 - \frac{1}{4}R_{ab}{}^{cd}M^{ab}M_{cd}. \quad (4.4)$$

The quantity u is defined in (3.36) and

$$T_{AB}{}^C = C_{AB}{}^C + \Omega_{AB}{}^C - (-)^{AB}\Omega_{BA}{}^C = (-)^{M(N+B)}E_A{}^ME_B{}^NT_{MN}{}^C \quad (4.5)$$

is the supertorsion. Note that $T_{ab}{}^a = C_{ab}{}^a$ and $T_{ab}{}^{\dot{a}} = C_{ab}{}^{\dot{a}}$. We see that relation (4.1) does not form the first-class algebra.

Let us begin with modifications of the constraints. We consider the quantity $\ell'_0 \approx 0$ as the first step modification $\ell_0 \rightarrow \ell'_0$ of the constraint $\ell_0 \approx 0$ and check a Poisson bracket of this new constraint with the constraints $\ell \approx 0$ and $\tilde{\ell} \approx 0$.

So let us consider the Poisson brackets of the quantity (4.4) with ℓ and $\tilde{\ell}$. The result of calculations has the form

$$\{\ell'_0, \ell\} = 2(\xi\sigma^a\bar{\xi})\mathcal{P}^b R_{ab}{}^{cd}M_{cd} - \frac{1}{4}(\xi\sigma^h\bar{\xi})D_h R_{ab}{}^{cd}M^{ab}M_{cd} + 2(\xi\sigma^a\bar{\xi})\mathcal{P}^b T_{ab}{}^C \mathcal{P}_C, \quad (4.6)$$

$$\{\ell'_0, \tilde{\ell}\} = 2(\bar{\pi}\tilde{\sigma}^a\pi)\mathcal{P}^b R_{ab}{}^{cd}M_{cd} - \frac{1}{4}(\bar{\pi}\tilde{\sigma}^h\pi)D_h R_{ab}{}^{cd}M^{ab}M_{cd} + 2(\bar{\pi}\tilde{\sigma}^a\pi)\mathcal{P}^b T_{ab}{}^C \mathcal{P}_C, \quad (4.7)$$

where

$$D_h R_{ab}{}^{cd} = \partial_h R_{ab}{}^{cd} + \Omega_{ha}{}^g R_{gb}{}^{cd} + \Omega_{hb}{}^g R_{ag}{}^{cd} + \Omega_h{}^c{}_g R_{ab}{}^{gd} + \Omega_h{}^d{}_g R_{ab}{}^{cg} \quad (4.8)$$

and $\partial_a = E_a{}^M \partial_M$. We see that the quantities ℓ'_0 , ℓ and $\tilde{\ell}$ defined in (4.4), (3.34) and (3.35) do not form a closed algebra in the general case. To obtain a closed algebra, we should impose some restrictions on supergeometry and perhaps make further modifications of the constraints.

As we pointed out, the modifications of the constraints should not change the number of first-class bosonic constraints. Let us pay attention that the initial constraints ℓ (3.34) and $\tilde{\ell}$ (3.35) contain the terms that are linear in $\mathcal{P}_a$. Modifying these constraints to include higher powers of $\mathcal{P}_a$ will almost inevitably lead to an increase in the number of constraints since the algebra of the constraints (4.6), (4.7) cannot be closed only on the constraints ℓ_0 , ℓ , $\tilde{\ell}$. To avoid the emergence of additional constraints, we should preserve linear dependence of expressions on $\mathcal{P}_a$. But ℓ and $\tilde{\ell}$ depend on $\mathcal{P}_a$ through the expressions $\mathcal{P}_a(\xi\sigma^a\bar{\xi})$ and $\mathcal{P}_a(\bar{\pi}\tilde{\sigma}^a\pi)$. To preserve only such linear in $\mathcal{P}_a$ expressions at modifications of the constraints, we are forced to impose restrictions on the background supergeometry.

Let us consider the right-hand sides of relations (4.1), (4.6) and (4.7) and find out under what conditions these relations will be closed when the constraints ℓ'_0 , ℓ , $\tilde{\ell}$ are satisfied. Taking into account the equalities

$$(\xi\sigma^a\bar{\xi})M_{ab} = -\frac{1}{2}(\xi\sigma_b\bar{\xi})K, \quad (\bar{\pi}\tilde{\sigma}^a\pi)M_{ab} = \frac{1}{2}(\bar{\pi}\tilde{\sigma}_b\pi)K, \quad (4.9)$$

we see that the needed terms $\mathcal{P}_a(\xi\sigma^a\bar{\xi})$ and $\mathcal{P}_a(\bar{\pi}\tilde{\sigma}^a\pi)$ will appear on the right-hand sides of (4.1), (4.6) and (4.7) only in the case when the components $R_{ab}{}^{cd}$ of the curvature tensor of background geometry are taken in the form

$$R_{ab}{}^{cd} = k(\delta_a^c\delta_b^d - \delta_a^d\delta_b^c), \quad (4.10)$$

where k is a constant. As is well known, relation (4.10) defines the constant curvature space-time. At zero, positive or negative k we get the Minkowski, de Sitter, and anti-de Sitter spaces, respectively.

As is well known, only the anti-de Sitter space is consistent with conventional supersymmetry [44] (also see, e.g., [43]).⁹ Therefore, below we will consider only the anti-de Sitter space with

$$k = -\kappa^2, \quad (4.11)$$

where κ is a real number. In this case nonvanishing components of the curvature and the torsion tensors take the form [43,44]:

$$\begin{aligned} R_{ab}{}^{cd} &= -2\kappa^2\delta_{[a}^c\delta_{b]}^d, & R_{a\dot{b}}{}^{\gamma\delta} &= -4\kappa\delta_{(a}^{\gamma}\delta_{\dot{b})}^{\delta}, & R_{\dot{a}\dot{b}}{}^{\dot{\gamma}\dot{\delta}} &= 4\kappa\delta_{(\dot{a}}^{\dot{\gamma}}\delta_{\dot{b})}^{\dot{\delta}}, \\ T_{a\dot{b}}{}^c &= -2i(\sigma^c)_{a\dot{b}}, & T_{a\dot{b}}{}^{\dot{\gamma}} &= \frac{i}{2}\kappa(\sigma_a)_{\dot{b}}^{\dot{\gamma}}, & T_{a\dot{b}}{}^{\gamma} &= -\frac{i}{2}\kappa(\sigma_a)_{\dot{b}}^{\gamma}. \end{aligned} \quad (4.12)$$

These quantities satisfy the reality conditions:

$$(R_{ab}{}^{cd})^* = R_{ab}{}^{cd}, \quad (R_{a\dot{b}}{}^{\gamma\delta})^* = -R_{\dot{a}b}{}^{\dot{\gamma}\dot{\delta}}, \quad (T_{a\dot{b}}{}^c)^* = -T_{\dot{a}b}{}^c, \quad (T_{a\dot{b}}{}^{\dot{\gamma}})^* = -T_{a\dot{b}}{}^{\dot{\gamma}}. \quad (4.13)$$

⁸ These relations are derived with the help of the identity $(\sigma^{[a})_{a\dot{a}}(\tilde{\sigma}^{b]})_{\dot{b}b} = -(\sigma^{ab})_a{}^{\gamma}\epsilon_{\gamma\dot{\beta}}\epsilon_{\dot{a}\dot{\beta}} - \epsilon_{\dot{a}\dot{\gamma}}(\tilde{\sigma}^{ab})^{\dot{\gamma}}{}_{\dot{\beta}}\epsilon_{a\beta}$

⁹ The formulation of supersymmetry models on de Sitter space requires some modification of the superalgebra (see, e.g., [50–53] and references therein.

We emphasize that in the curved superspace the equalities $T_{\alpha\beta}{}^c = C_{\alpha\beta}{}^c$, $T_{a\beta}{}^{\dot{\gamma}} = C_{a\beta}{}^{\dot{\gamma}}$, $T_{a\dot{\beta}}{}^{\gamma} = C_{a\dot{\beta}}{}^{\gamma}$ hold, which reflects the fact that there are no such corresponding nonvanishing components of spin connections. Besides, the curvature tensor (4.10) of constant curvature spaces is a covariantly constant:

$$D_{\dot{h}} R_{ab}{}^{cd} = 0. \quad (4.14)$$

As result, in anti-de Sitter superspace (4.12) we get the following Poisson bracket of the constraints ℓ and $\tilde{\ell}$

$$\{\ell, \tilde{\ell}\} = -K\ell_0'', \quad (4.15)$$

where we have used the relations

$$-\frac{1}{4} R_{ab}{}^{cd} M^{ab} M_{cd} = \frac{1}{2} k(N^2 + \tilde{N}^2), \quad -\frac{1}{4} (\xi\sigma^{ab}\pi + \bar{\pi}\tilde{\sigma}^{ab}\bar{\xi}) R_{ab}{}^{cd} M_{cd} = \frac{1}{2} k K u, \quad (4.16)$$

and the second stage modified initial constraint has been introduced

$$\ell_0'' := \ell_0' - \frac{1}{2} k u^2 = \ell_0 - \frac{1}{4} R_{ab}{}^{cd} M^{ab} M_{cd} - \frac{1}{2} k u^2 = \ell_0 + k N \tilde{N} = P_a P^a + k N \tilde{N}. \quad (4.17)$$

Let us consider the Poisson bracket of the constraint ℓ_0'' (4.17) with the constraints (3.34), (3.35). The direct calculations lead to the following result:

$$\{\ell_0'', \ell\} = -2kK(\ell + \mu), \quad \{\ell_0'', \tilde{\ell}\} = 2kK(\tilde{\ell} + \mu). \quad (4.18)$$

This algebra is not closed due to the presence of μ -dependent terms in the right-hand side of (4.18). Therefore, the bosonic constraints still need to be modified in curved superspace. This modification can be made exactly as in the purely bosonic case [23] by adding extra terms with operators N and $\tilde{N}$.

Let us begin with the modification of the constraint (4.17) requiring that the modified constraint commutes with ℓ and $\tilde{\ell}$. Using the Poisson brackets

$$\{N, \ell\} = \{\tilde{N}, \ell\} = -(\ell + \mu), \quad \{N, \tilde{\ell}\} = \{\tilde{N}, \tilde{\ell}\} = \tilde{\ell} + \mu \quad (4.19)$$

and (4.18), we see that the quantity, which is equal to the sum of ℓ_0'' plus the additional term

$$-\frac{k}{2} K^2 \quad \text{or} \quad -2kN\tilde{N} \quad \text{or} \quad -k(N^2 + \tilde{N}^2), \quad (4.20)$$

has zero Poisson brackets with ℓ and $\tilde{\ell}$. However, at the constraint $U = N - \tilde{N} \approx 0$, the following equalities take place:

$$-\frac{k}{2} K^2 = -2kN\tilde{N} = -k(N^2 + \tilde{N}^2), \quad (4.21)$$

which allows us to reduce the number of variants in (4.20). We choose the second option and define the quantity

$$\Downarrow_{0,\gamma} := \ell_0'' - 2kN\tilde{N} + \gamma = P_a P^a - kN\tilde{N} + \gamma \approx 0 \quad (4.22)$$

as a true modified constraint with an arbitrary constant parameter γ which will be fixed later. Let us emphasize once again the important property obtained, the constraint (4.22) has zero Poisson brackets with ℓ and $\tilde{\ell}$:

$$\{\Downarrow_{0,\gamma}, \ell\} = \{\Downarrow_{0,\gamma}, \tilde{\ell}\} = 0. \quad (4.23)$$

We will show that this quantity can be associated with true constraint $\Downarrow_0$ at some fixed γ .

Let us turn to the constraints $\ell \approx 0$, $\tilde{\ell} \approx 0$. Relation (4.18) shows that their Poisson bracket can not reproduce the quantity $\Downarrow_{0,\gamma}$ at any γ . Therefore, if we desire to obtain the true constraints $\Downarrow$ and $\tilde{\Downarrow}$ from the quantities $\ell \approx 0$ and $\tilde{\ell} \approx 0$, the latter should be modified in such a way that their Poisson bracket is equal to a linear combination of new constraints, unlike the Poisson bracket (4.18).

The desired modification is realized by adding the terms containing $N\tilde{N}$ to the constraints $\ell \approx 0$, $\tilde{\ell} \approx 0$. As a result, we arrive to the following functions:

$$\Downarrow_{\beta_1} := \ell + \beta_1 N\tilde{N} \approx 0, \quad \tilde{\Downarrow}_{\beta_2} := \tilde{\ell} + \beta_2 N\tilde{N} \approx 0, \quad (4.24)$$

where β_1 and β_2 are some arbitrary constants. In the general case, these constants are not equal to each other, $\beta_1 \neq \beta_2$, since the constraints (4.24) are real and are not related one to another.

Now we require that the quantities $\Downarrow_{0,\gamma}$, $\Downarrow_{\beta_1}$, $\tilde{\Downarrow}_{\beta_2}$ form the closed algebra. We will show that this requirement completely fixes the constants γ , β_1 , β_2 .

Since $\{M_{ab}, K\} = 0$, the constraint (4.22) has zero Poisson brackets with constraints (4.24): $\{\Downarrow_{0,\gamma}, \Downarrow_{\beta_1}\} = \{\Downarrow_{0,\gamma}, \tilde{\Downarrow}_{\beta_2}\} = 0$. The Poisson bracket of the quantities (4.24) equals

$$\{\Downarrow_{\beta_1}, \tilde{\Downarrow}_{\beta_2}\} = -K\Downarrow_0 + K(\beta_2\Downarrow_{\beta_1} + \beta_1\tilde{\Downarrow}_{\beta_2}) - 2(k + \beta_1\beta_2)KN\tilde{N} + (\gamma + \mu(\beta_1 + \beta_2))K. \quad (4.25)$$

To close off this algebra, the third and fourth terms on the right-hand side must be equal to zero. This leads to the following conditions on the constants γ , β_1 and β_2 :

$$\beta_1\beta_2 = -k, \quad \gamma = -\mu(\beta_1 + \beta_2). \quad (4.26)$$

Moreover, we can consider the case when $\beta_1 = \beta_2$. Then, the conditions (4.26) yield

$$\beta_1 = \beta_2 = \kappa, \quad \gamma = -2\kappa\mu. \quad (4.27)$$

As a result, we obtain that the bosonic constraints (4.22), (4.24) have the form

$$\Downarrow_0 = \mathcal{P}_a \mathcal{P}^a + \kappa^2 N \bar{N} - 2\kappa\mu \approx 0 \quad (4.28)$$

$$\Downarrow = (\xi \sigma^a \bar{\xi}) \mathcal{P}_a + \kappa N \bar{N} - \mu \approx 0, \quad (4.29)$$

$$\tilde{\Downarrow} = (\bar{\pi} \tilde{\sigma}^a \pi) \mathcal{P}_a + \kappa N \bar{N} - \mu \approx 0. \quad (4.30)$$

The fourth bosonic constraint is the constraint (3.36) having the form

$$\mathcal{U} = N - \bar{N} \approx 0. \quad (4.31)$$

The nonvanishing Poisson bracket of the constraint (4.28)–(4.31) is

$$\{\Downarrow, \tilde{\Downarrow}\} = -K \Downarrow_0 + \kappa K (\Downarrow + \tilde{\Downarrow}). \quad (4.32)$$

Note that the constraints (4.28)–(4.30) in the AdS space can be represented in the form similar to the corresponding constraints (2.7)–(2.9) in flat space. For this we introduce the vector quantity

$$\Pi_a := \mathcal{P}_a - \frac{\kappa}{2} (\xi \sigma_a \bar{\xi} + \bar{\pi} \tilde{\sigma}_a \pi). \quad (4.33)$$

Using vector (4.33), the constraints (4.29), (4.30) are represented as

$$\Downarrow = \Pi_a (\xi \sigma^a \bar{\xi}) - \mu \approx 0, \quad (4.34)$$

$$\tilde{\Downarrow} = \Pi_a (\bar{\pi} \tilde{\sigma}^a \pi) - \mu \approx 0. \quad (4.35)$$

Moreover, from the explicit form (4.33) for the vector Π_a we have the equality

$$\Pi_a \Pi^a = \Downarrow_0 - \kappa (\Downarrow + \tilde{\Downarrow}). \quad (4.36)$$

That is, the set of constraints (4.28)–(4.30) and the set of constraints consisting of both the constraint

$$\hat{\Downarrow}_0 = \Pi_a \Pi^a \approx 0 \quad (4.37)$$

and the constraints (4.35), (4.35) are equivalent.

The form of the constraints (4.37)–(4.35) in the AdS space is similar to the form of the constraints (2.7)–(2.9) in flat space. The constraints (4.35)–(4.37) turn into the constraints (2.7)–(2.9) through the replacement $p_a \leftrightarrow \Pi_a$.

One can see that the transition from the system with bosonic constraints (3.33)–(3.35) to the system with bosonic constraints (4.28)–(4.30) is achieved in the case of AdS superspace by adding the extra term

$$L_{extra} = \frac{\kappa}{2} \dot{z}^M E_M^a [(\xi \sigma_a \bar{\xi}) + (\bar{\pi} \tilde{\sigma}_a \pi)] \quad (4.38)$$

to the Lagrangian (3.3). After the indicated modification of the three bosonic constraints, we arrive at the following total Lagrangian:

$$L_{tot} = L + L_{extra}. \quad (4.39)$$

This Lagrangian has the same fourth bosonic constraint (4.31) and the same fermionic constraints (3.38). As we will see in the next section, no further modification of the fermionic constraints is required and the Lagrangian (4.39) is the final Lagrangian of the system under consideration.

4.2. Structure of the fermionic constraints

Modified bosonic constraints must have zero Poisson brackets (in the weak sense) with fermionic quantities (3.38): $\mathcal{D}_\alpha$ and $\bar{\mathcal{D}}_{\dot{\alpha}}$. This will be a necessary condition for the quantities $\mathcal{D}_\alpha \approx 0$ and $\bar{\mathcal{D}}_{\dot{\alpha}} \approx 0$ to be fermionic constraints that do not require additional modification of bosonic constraints.

Using (B.1)–(B.6), we obtain that the Poisson brackets of the constraints (4.28)–(4.30) with (3.38) have the form (here the identities $\Omega_A^{ab} \mathcal{P}_a \mathcal{P}_b \equiv 0$ are taken into account):

$$\{\Downarrow_0, \mathcal{D}_\beta\} = -i \mathcal{P}^a (R_{a\beta}{}^{cd} M_{cd} + 2 T_{a\beta}{}^C \mathcal{P}_C) + 2 \mathcal{P}^a \Omega_{a\beta}{}^\gamma \mathcal{D}_\gamma, \quad (4.40)$$

$$\{\Downarrow, \mathcal{D}_\beta\} = -i (\xi \sigma^a \bar{\xi}) \left(\frac{1}{2} R_{a\beta}{}^{cd} M_{cd} + T_{a\beta}{}^C \mathcal{P}_C \right) + (\xi \sigma^a \bar{\xi}) \Omega_{a\beta}{}^\gamma \mathcal{D}_\gamma, \quad (4.41)$$

$$\{\tilde{\Downarrow}, \mathcal{D}_\beta\} = -i (\bar{\pi} \tilde{\sigma}^a \pi) \left(\frac{1}{2} R_{a\beta}{}^{cd} M_{cd} + T_{a\beta}{}^C \mathcal{P}_C \right) + (\bar{\pi} \tilde{\sigma}^a \pi) \Omega_{a\beta}{}^\gamma \mathcal{D}_\gamma \quad (4.42)$$

and c.c. brackets.¹⁰ In AdS superspace, the nonvanishing components of the supercurvature and supertension are given in (4.12). Therefore, in this case, the Poisson brackets (4.40)–(4.42) take the form

$$\begin{aligned} \{\downarrow_0, \mathcal{D}_\beta\} &= i\kappa P_a \sigma_{\beta\gamma}^a \tilde{\mathcal{D}}^\gamma + 2 P^a \Omega_{a\beta}{}^\gamma \mathcal{D}_\gamma, \\ \{\downarrow, \mathcal{D}_\beta\} &= -i\kappa \xi_\beta \tilde{\xi}_\gamma \tilde{\mathcal{D}}^\gamma + (\xi^\alpha \tilde{\xi}) \Omega_{a\beta}{}^\gamma \mathcal{D}_\gamma, \\ \{\downarrow, \tilde{\mathcal{D}}_\beta\} &= -i\kappa \pi_\beta \tilde{\pi}_\gamma \tilde{\mathcal{D}}^\gamma + (\tilde{\pi} \tilde{\sigma}^a \pi) \Omega_{a\beta}{}^\gamma \mathcal{D}_\gamma \end{aligned} \quad (4.43)$$

and c.c. brackets. Thus, after modification of the bosonic constraints by adding extra terms with the operators N and $\tilde{N}$, the new bosonic constraints (4.28)–(4.30) have vanishing (in the weak sense) brackets with the fermionic constraints $\mathcal{D}_\alpha \approx 0$ and $\tilde{\mathcal{D}}_{\dot{\alpha}} \approx 0$. Besides, the constraint $\mathcal{U} \approx 0$ defined in (4.31) has also zero brackets with $\mathcal{D}_\alpha$ and $\tilde{\mathcal{D}}_{\dot{\alpha}}$:

$$\{\mathcal{U}, \mathcal{D}_\alpha\} = 0, \quad \{\mathcal{U}, \tilde{\mathcal{D}}_{\dot{\alpha}}\} = 0. \quad (4.44)$$

Hence, the fermionic constraints under consideration $\mathcal{D}_\alpha \approx 0$ and $\tilde{\mathcal{D}}_{\dot{\alpha}} \approx 0$ commute (in the weak sense) with all bosonic constraints in terms of the Poisson brackets:

- all bosonic constraints (4.28)–(4.30), (4.31) are of the first class in the supersymmetric system under consideration;
- in the flat limit, the constraints $\mathcal{D}_\alpha \approx 0$, $\tilde{\mathcal{D}}_{\dot{\alpha}} \approx 0$ (3.38) transform into the flat fermionic constraints $D_\alpha \approx 0$, $\tilde{D}_{\dot{\alpha}} \approx 0$ (2.6).

Now let us analyze the classes of the fermionic constraints $\mathcal{D}_\alpha \approx 0$ and $\tilde{\mathcal{D}}_{\dot{\alpha}} \approx 0$. Using (B.2)–(B.4), we obtain that the Poisson brackets of the fermionic constraints $\mathcal{D}_\alpha \approx 0$ and $\tilde{\mathcal{D}}_{\dot{\alpha}} \approx 0$ in AdS space have the form (on the surface of the constraints $\mathcal{D}_\alpha \approx 0$ and $\tilde{\mathcal{D}}_{\dot{\alpha}} \approx 0$)

$$\begin{pmatrix} \{\mathcal{D}_\alpha, \mathcal{D}^\beta\} & \{\mathcal{D}_\alpha, \tilde{\mathcal{D}}_\beta\} \\ \{\tilde{\mathcal{D}}_{\dot{\alpha}}, \mathcal{D}^\beta\} & \{\tilde{\mathcal{D}}_{\dot{\alpha}}, \tilde{\mathcal{D}}_\beta\} \end{pmatrix} = -2i C_{\underline{A}}{}^{\underline{B}}, \quad (4.45)$$

where

$$C_{\underline{A}}{}^{\underline{B}} := \begin{pmatrix} -2i\kappa M_\alpha{}^\beta & (\sigma^c)_{\alpha\beta} \mathcal{P}_c \\ (\tilde{\sigma}^c)^{\dot{\alpha}\beta} \mathcal{P}_c & 2i\kappa \tilde{M}^{\dot{\alpha}}{}_{\dot{\beta}} \end{pmatrix}. \quad (4.46)$$

In (4.45), (4.46) the index $\underline{A}$ is the 4-component Dirac-spinor index:

$$X_{\underline{A}} = \begin{pmatrix} X_\alpha \\ X_{\dot{\alpha}} \end{pmatrix}, \quad X^{\underline{A}} = (X^\alpha, X_{\dot{\alpha}}); \quad (4.47)$$

and $M_{\alpha\beta}$ and $\tilde{M}_{\dot{\alpha}\dot{\beta}}$ are defined in (3.28). As a result, the 4×4 matrix equality (4.45) is represented as

$$\{\mathcal{D}_{\underline{A}}, \mathcal{D}^{\underline{B}}\} = -2i C_{\underline{A}}{}^{\underline{B}}. \quad (4.48)$$

Note that after using the equalities (see (3.27))

$$M_{\alpha\beta} = \frac{1}{2} M_{ab} (\sigma^{ab})_{\alpha\beta}, \quad \tilde{M}^{\dot{\alpha}\dot{\beta}} = -\frac{1}{2} M_{ab} (\tilde{\sigma}^{ab})^{\dot{\alpha}\dot{\beta}}, \quad (4.49)$$

the matrix (4.46) looks like

$$C = \mathcal{P}_a \gamma^a - i M_{ab} \Sigma^{ab}, \quad (4.50)$$

where

$$(\gamma^a)_{\underline{A}}{}^{\underline{B}} = \begin{pmatrix} 0 & (\sigma^a)_{\alpha\beta} \\ (\tilde{\sigma}^a)^{\dot{\alpha}\beta} & 0 \end{pmatrix} \quad (4.51)$$

are the Dirac matrices and

$$\Sigma^{ab} = -\frac{1}{4} [\gamma^a, \gamma^b], \quad (\Sigma^{ab})_{\underline{A}}{}^{\underline{B}} = \begin{pmatrix} (\sigma^{ab})_{\alpha}{}^\beta & 0 \\ 0 & (\tilde{\sigma}^{ab})^{\dot{\alpha}}{}_{\dot{\beta}} \end{pmatrix} \quad (4.52)$$

are the Lorentz generators in the Dirac-spinor representation.

To calculate the determinant of the matrix (4.45), we use the relation

$$\det \begin{pmatrix} A_\alpha{}^\beta & B_{\alpha\beta} \\ B^{\dot{\alpha}\beta} & \tilde{A}^{\dot{\alpha}}{}_{\dot{\beta}} \end{pmatrix} = \frac{1}{4} (B_{\alpha\beta} B^{\beta\alpha})^2 + \frac{1}{4} A^{\alpha\beta} A_{\alpha\beta} \tilde{A}^{\dot{\alpha}\dot{\beta}} \tilde{A}_{\dot{\alpha}\dot{\beta}} - A_\alpha{}^\beta B_{\beta\gamma} \tilde{A}^{\dot{\gamma}}{}_{\dot{\delta}} B^{\delta\alpha}, \quad (4.53)$$

¹⁰ Let us emphasize the presence of the minus sign in the superconnection matrix (see, e.g., [43]):

$$\Omega_{AB}{}^C = \begin{pmatrix} \Omega_{Ab}{}^c & 0 & 0 \\ 0 & \Omega_{A\beta}{}^\gamma & 0 \\ 0 & 0 & -\Omega_A{}^\beta{}_{\dot{\gamma}} \end{pmatrix}.$$

where $A_{\alpha\beta} = \epsilon_{\beta\gamma} A_{\alpha}{}^{\gamma}$, $A^{\alpha\beta} = \epsilon^{\alpha\gamma} A_{\gamma}{}^{\beta}$, $\bar{A}^{\dot{\alpha}\dot{\beta}} = \epsilon^{\dot{\beta}\dot{\gamma}} \bar{A}^{\dot{\alpha}}{}_{\dot{\gamma}}$, $\bar{A}_{\dot{\alpha}\dot{\beta}} = \epsilon_{\dot{\alpha}\dot{\gamma}} \bar{A}^{\dot{\gamma}}{}_{\dot{\beta}}$, $B^{\dot{\alpha}\dot{\beta}} = \epsilon^{\dot{\alpha}\dot{\gamma}} \epsilon^{\dot{\beta}\dot{\delta}} B_{\dot{\delta}\dot{\gamma}}$ and all matrix components are c -numbers. Using (4.53) and the identity

$$(\xi\sigma^a\bar{\pi})\mathcal{P}_a(\pi\sigma^b\bar{\xi})\mathcal{P}_b = (\xi\sigma^a\bar{\xi})\mathcal{P}_a(\pi\sigma^b\bar{\pi})\mathcal{P}_b + N\bar{N}\mathcal{P}^a\mathcal{P}_a,$$

we obtain that the determinant of the Poisson brackets matrix (4.45) has the form

$$\det C = (\mathcal{P}^a\mathcal{P}_a)^2 + |\kappa|^4 N^2 \bar{N}^2 - 4|\kappa|^2 (\xi\sigma^a\bar{\xi})\mathcal{P}_a(\pi\sigma^b\bar{\pi})\mathcal{P}_b - 2|\kappa|^2 N\bar{N}\mathcal{P}^a\mathcal{P}_a. \quad (4.54)$$

Using the expressions for $\mathcal{P}^a\mathcal{P}_a$, $(\xi\sigma^a\bar{\xi})\mathcal{P}_a(\pi\sigma^b\bar{\pi})\mathcal{P}_b$ from (4.28)–(4.30), one obtains

$$\det C = (\mathcal{P}_0)^2 - 4\kappa(\kappa N\bar{N} - \mu)[\mathcal{P}_0 - \kappa(\mathcal{P} + \bar{\mathcal{P}})] - 4\kappa^2 \mathcal{P}\bar{\mathcal{P}}. \quad (4.55)$$

Therefore, on the surface of bosonic constraints, the determinant of the matrix constructed from the Poisson brackets of fermionic constraints is equal to zero:

$$\det C \approx 0. \quad (4.56)$$

Thus, in the case under consideration, four constraints $\mathcal{D}_\alpha \approx 0$ and $\bar{\mathcal{D}}_{\dot{\alpha}} \approx 0$ form a mixture of first- and second-class constraints. Moreover, of these four fermionic constraints, at least one constraint is the first-class one.

4.3. Division of the fermionic constraints into first- and second-class constraints

In this subsection we will consider the division of the four fermionic constraints $\mathcal{D}_\alpha \approx 0$ and $\bar{\mathcal{D}}_{\dot{\alpha}} \approx 0$ into the first- and second-class constraints. First of all, let us find the numbers of separate first- and separate second-class constraints that are present in the above fermionic constraints.

Considering the matrix C (4.50), one can see that the matrix $\gamma^0 C$ is Hermitian and therefore is diagonalized by a unitary transformation. The corresponding characteristic equation that determines the eigenvalues and rank of the matrix $\gamma^0 C$ has the form

$$\det(\gamma^0 C - \lambda \mathbf{1}_4) = 0, \quad (4.57)$$

where the matrix $\mathbf{1}_4$ is the identity 4×4 matrix and the number λ is the eigenvalue of the matrix C . Eq. (4.57) is equivalent to the equation

$$\det(C - \lambda \gamma^0) = 0. \quad (4.58)$$

Eq. (4.58) is obtained from Eq. (4.56) by substituting

$$\mathcal{P}_0 \rightarrow \mathcal{P}_0 - \lambda. \quad (4.59)$$

Therefore, one gets $\mathcal{P}_0 \rightarrow \mathcal{P}_0 + 2\mathcal{P}_0\lambda - \lambda^2$, $\mathcal{P} \rightarrow \mathcal{P} + (\xi\sigma_0\bar{\xi})\lambda$, $\bar{\mathcal{P}} \rightarrow \bar{\mathcal{P}} + (\pi\sigma_0\bar{\pi})\lambda$. Then, using (4.55), we obtain

$$\det(C - \lambda \gamma^0) = \det C + a\lambda + b\lambda^2 - 4\mathcal{P}_0\lambda^3 + \lambda^4, \quad (4.60)$$

where

$$a = 4\mathcal{P}_0\mathcal{P}_0 - 4\kappa^2(\pi\sigma_0\bar{\pi})\mathcal{P} - 4\kappa^2(\xi\sigma_0\bar{\xi})\bar{\mathcal{P}} + 8\kappa(\mu - \kappa N\bar{N})\Pi_0, \quad (4.61)$$

$$b = 4(\mathcal{P}_0)^2 + 2\mathcal{P}_0 + \kappa(\kappa N\bar{N} - \mu) - (\xi\sigma_0\bar{\xi})(\pi\sigma_0\bar{\pi}), \quad (4.62)$$

where Π_0 is defined in (4.33).

On the constraint surface the quantity (4.61) takes the form $a = 8\kappa(\mu - \kappa N\bar{N})\Pi_0$. Further we assume that $(\mu - \kappa N\bar{N}) \neq 0$. Otherwise we would have one more bosonic constraint $N\bar{N} - \mu/\kappa \approx 0$, which, together with the existing constraint (4.31), would completely fix the values of N and $\bar{N}$, which corresponds to a particle of finite spin (helicity) (see, e.g., the consideration of such systems in [54]). Also note that on the constraint surface (4.37) the vector Π_a is light-like. Therefore, $\Pi_0 \neq 0$. As result, $a \neq 0$. Therefore, the Hermitian matrix $\gamma^0 C$ has only one zero eigenvalue. Thus, we have one first-class constraint and three second-class constraints among four fermionic constraints $\mathcal{D}_\alpha \approx 0$ and $\bar{\mathcal{D}}_{\dot{\alpha}} \approx 0$.

We now proceed to explicitly divide the above four fermionic constraints into one first-class constraint and three second-class ones. The division procedure is in principle similar to what was done in the flat case (see (2.17), (2.18)). But now we use the identities

$$\delta_\alpha^\beta = \frac{1}{\mu} [\bar{\xi}_\alpha (\bar{\xi}\bar{\Pi})^\beta + (\Pi\bar{\xi})_\alpha \bar{\xi}^\beta], \quad \delta_{\dot{\alpha}}^{\dot{\beta}} = \frac{1}{\mu} [\bar{\xi}_{\dot{\alpha}} (\bar{\Pi}\bar{\xi})^{\dot{\beta}} + (\bar{\xi}\Pi)_{\dot{\alpha}} \bar{\xi}^{\dot{\beta}}], \quad (4.63)$$

which are a curve superspace generalization of flat identities (2.22) and are fulfilled on the constraint surface (4.34) and (4.37).

Taking into account identities (4.63), we consider two constraints

$$\mathcal{G} := \xi^\alpha \mathcal{D}_\alpha \approx 0, \quad \bar{\mathcal{G}} := \bar{\mathcal{D}}_{\dot{\alpha}} \bar{\xi}^{\dot{\alpha}} \approx 0, \quad (\mathcal{G})^* = -\bar{\mathcal{G}} \quad (4.64)$$

and two constraints

$$\mathcal{F} := \Pi^b \bar{\xi}_{\dot{a}} \sigma_b^{\dot{a}\alpha} \mathcal{D}_\alpha \approx 0, \quad \bar{\mathcal{F}} := \bar{\mathcal{D}}_{\dot{a}} \sigma_b^{\dot{a}\alpha} \xi_a \Pi^b \approx 0, \quad (\mathcal{F})^* = -\bar{\mathcal{F}}. \quad (4.65)$$

The nonvanishing Poisson brackets of the constraints (4.64), (4.65) in the superspace with the supercurvature and supertorsion (4.12) have the form:

$$\{\mathcal{G}, \bar{\mathcal{G}}\} = -2i(\xi\mathcal{P}\bar{\xi}), \quad \{\mathcal{G}, \mathcal{G}\} = 0, \quad \{\bar{\mathcal{G}}, \bar{\mathcal{G}}\} = 0, \quad (4.66)$$

$$\{\mathcal{G}, F\} = 2\kappa N(\xi\Pi\bar{\xi}) + \kappa\mathcal{G}\bar{\mathcal{G}} - \kappa\bar{N}\mathcal{D}^a\mathcal{D}_a, \quad (4.67)$$

$$\{\bar{\mathcal{G}}, \bar{F}\} = -2\kappa\bar{N}(\xi\Pi\bar{\xi}) - \kappa\mathcal{G}\bar{\mathcal{G}} + \kappa N\bar{\mathcal{D}}_a\bar{\mathcal{D}}^a, \quad (4.68)$$

$$\{\mathcal{G}, \bar{F}\} = 2i\kappa N(\xi\Pi\bar{\pi}) + \kappa\mathcal{G}\bar{\mathcal{G}}, \quad (4.69)$$

$$\{\bar{\mathcal{G}}, F\} = 2i\kappa\bar{N}(\pi\Pi\bar{\xi}) + \kappa\mathcal{G}\bar{\mathcal{G}}, \quad (4.70)$$

$$\{F, F\} = -2\kappa\bar{\mathcal{G}}F + 2\kappa(\pi^a\mathcal{D}_a)F + 4\kappa(\pi\Pi\bar{\xi})(\xi\Pi\bar{\xi}), \quad (4.71)$$

$$\{\bar{F}, \bar{F}\} = -2\kappa\mathcal{G}\bar{F} + 2\kappa(\bar{\pi}^a\bar{\mathcal{D}}_a)\bar{F} - 4\kappa(\xi\Pi\bar{\pi})(\xi\Pi\bar{\xi}), \quad (4.72)$$

$$\{F, \bar{F}\} = -2i(\xi\Pi\bar{\mathcal{P}}\Pi\bar{\xi}) + \kappa(\mathcal{D}^a\mathcal{D}_a - \bar{\mathcal{D}}_a\bar{\mathcal{D}}^a)(\xi\Pi\bar{\xi}) + \kappa(\mathcal{D}\Pi\bar{\mathcal{D}})u. \quad (4.73)$$

Taking into account the equality $(\xi\Pi\bar{\mathcal{P}}\Pi\bar{\xi}) = (\xi\mathcal{P}\bar{\xi})\Pi^a\Pi_a - 2(\xi\Pi\bar{\xi})\Pi^a\mathcal{P}_a \approx -2\kappa\mu^2$ and also (A.7), we obtain that on the constraint surface the Poisson brackets of the fermionic constraints have the form

$$\{\mathcal{G}, \bar{\mathcal{G}}\} \approx -2i(\mu - \kappa N\bar{N}), \quad \{\mathcal{G}, \mathcal{G}\} \approx 0, \quad \{\bar{\mathcal{G}}, \bar{\mathcal{G}}\} \approx 0, \quad (4.74)$$

$$\begin{aligned} \{\mathcal{G}, F\} &\approx 2\kappa\mu N, & \{\mathcal{G}, \bar{F}\} &\approx 2i\kappa\mu N, \\ \{\bar{\mathcal{G}}, F\} &\approx 2i\kappa\mu\bar{N}, & \{\bar{\mathcal{G}}, \bar{F}\} &\approx -2\kappa\mu\bar{N}, \end{aligned} \quad (4.75)$$

$$\{F, F\} \approx 4\bar{c}\kappa\mu^2, \quad \{\bar{F}, \bar{F}\} \approx -4c\kappa\mu^2, \quad \{F, \bar{F}\} \approx 4i\kappa\mu^2. \quad (4.76)$$

Using relations (4.75) and (4.76), we obtain that the constraint

$$F_1 := F + \bar{c}i\bar{F} \quad (4.77)$$

has the vanishing Poisson brackets with all fermionic constraints

$$\{\mathcal{G}, F_1\} \approx \{\bar{\mathcal{G}}, F_1\} \approx \{F, F_1\} \approx \{\bar{F}, F_1\} \approx 0 \quad (4.78)$$

and with itself (since this constraint is Grassmann odd, this is a non-trivial condition)

$$\{F_1, F_1\} \approx 0. \quad (4.79)$$

Therefore, expression (4.77) is the first-class fermionic constraint. The realness condition for this constraint looks like: $(F_1)^* = ciF_1$. When choosing $c = -i$, the constraint (4.77) is real in conventional form: $(F_1)^* = F_1$.

The remaining constraints, namely, the constraint

$$F_2 := F - \bar{c}i\bar{F} \quad (4.80)$$

and the constraints $\mathcal{G}$ and $\bar{\mathcal{G}}$, are the second-class constraints.

To obtain the explicit forms of the expressions $\mathcal{G}$ and $\bar{\mathcal{G}}$, we divide these constraints into real and imaginary parts as follows:

$$\mathcal{G}_1 := \mathcal{G} + ci\bar{\mathcal{G}}, \quad \mathcal{G}_2 := \mathcal{G} - ci\bar{\mathcal{G}}. \quad (4.81)$$

The Poisson brackets of the constraints (4.81) and (4.80) have the form

$$\{\mathcal{G}_1, \mathcal{G}_1\} \approx 4c(\mu - \kappa N\bar{N}), \quad \{\mathcal{G}_2, \mathcal{G}_2\} \approx -4c(\mu - \kappa N\bar{N}), \quad \{\mathcal{G}_1, \mathcal{G}_2\} \approx 0, \quad (4.82)$$

$$\{F_2, F_2\} \approx 16\bar{c}\kappa\mu^2, \quad \{\mathcal{G}_1, F_2\} \approx 0, \quad \{\mathcal{G}_2, F_2\} \approx 4\kappa\mu(N + \bar{N}). \quad (4.83)$$

The constraint $\mathcal{G}_1$ has splitted off and we clearly see that it is of the second-class.

For the remaining two constraints $\mathcal{G}_2$ and F_2 , the Poisson bracket matrix

$$\begin{pmatrix} \{\mathcal{G}_2, \mathcal{G}_2\} & \{\mathcal{G}_2, F_2\} \\ \{F_2, \mathcal{G}_2\} & \{F_2, F_2\} \end{pmatrix} \approx 4 \begin{pmatrix} -c(\mu - \kappa N\bar{N}) & \kappa\mu(N + \bar{N}) \\ \kappa\mu(N + \bar{N}) & 4\bar{c}\kappa\mu^2 \end{pmatrix} \quad (4.84)$$

has the determinant that is proportional to $\kappa\mu^3$ on the constraints surface and is non-zero. This shows that the constraints $\mathcal{G}_2$ and F_2 are the second-class ones.

As a result, the model of a continuous spin superparticle in AdS_4 superspace we have constructed is described by four bosonic first-class constraints $\hat{\mathbb{P}}_0 \approx 0$ (4.28), $\hat{\mathbb{P}} \approx 0$ (4.29), $\hat{\mathbb{P}} \approx 0$ (4.30), $U \approx 0$ (4.31) and four fermionic constraints $\mathcal{D}_a \approx 0$, $\bar{\mathcal{D}}_a \approx 0$ (3.38). The latter are a mixture of one first-class constraint $F_1 \approx 0$ (4.77) and three second-class constraints $F_2 \approx 0$ (4.80), $\mathcal{G}_1 \approx 0$, $\mathcal{G}_2 \approx 0$ (4.81).

Let us note the difference between the model of a continuous spin superparticle in flat superspace and in anti-de Sitter superspace. Despite the same type of bosonic constraints (there are four of them and they are all of the first class), the fermionic constraints have a different structure. In the flat case, four fermionic constraints $D_a \approx 0$, $\bar{D}_a \approx 0$ (2.6) form a mixture of two irreducible second-class constraints $G \approx 0$, $\bar{G} \approx 0$ (2.17) and two irreducible first-class constraints $F \approx 0$, $\bar{F} \approx 0$ (2.18). In AdS_4 superspace, the analogue of the flat constraints $D_a \approx 0$, $\bar{D}_a \approx 0$ (2.6) is the constraints $\mathcal{D}_a \approx 0$, $\bar{\mathcal{D}}_a \approx 0$ (3.38). But now only one of these constraints, namely, the real part $F_1 \approx 0$ (4.77), is the first-class constraint. The imaginary part $F_2 \approx 0$ (4.80) and $\mathcal{G}_1 \approx 0$, $\mathcal{G}_2 \approx 0$ (4.81) are the second-class constraints.

4.4. Fermionic κ -symmetry in curved superspace

As is well known, any first-class constraint is a generator of gauge transformations in phase space (see, e.g., [49]). In the case under consideration, we have only one first-class fermionic constraint (4.77). This constraint generates a local fermionic transformation of the curved phase superspace coordinates in the form

$$\delta z^M = \kappa \{F_1, z^M\}, \quad \delta \xi^\alpha = \kappa \{F_1, \xi^\alpha\}, \quad \delta \bar{\xi}^{\dot{\alpha}} = \kappa \{F_1, \bar{\xi}^{\dot{\alpha}}\}, \quad (4.85)$$

with κ being the only one (real in the case $c = -i$) anticommuting local transformation parameter. Similar transformations will be held for the momenta p_M , π_α , $\bar{\pi}_{\dot{\alpha}}$. Direct calculations with the use of the explicit expression for the first-class constraint (4.77) lead to the following κ -symmetry transformations for the continuous superparticle model in curved superspace:

$$\delta z^M = -\kappa [\bar{\xi}_{\dot{\alpha}} \tilde{\Pi}^{\dot{\alpha}\alpha} E_\alpha^M + \bar{c} i \xi^\alpha \Pi_{\alpha\dot{\alpha}} E^{\dot{\alpha}M}] - \kappa [(\bar{\xi} \tilde{\sigma}^a \mathcal{D}) + \bar{c} i (\xi \sigma^a \tilde{\mathcal{D}})] E_a^M, \quad (4.86)$$

$$\delta \xi^\alpha = i \kappa [(\bar{\xi} \tilde{\Pi})^\gamma \Omega_\gamma^{\alpha\beta} + \bar{c} i (\xi \Pi)_\gamma \Omega^{\gamma\alpha\beta}] \xi_\beta + \kappa [(\bar{\xi} \tilde{\sigma}^a \mathcal{D}) + \bar{c} i (\xi \sigma^a \tilde{\mathcal{D}})] \Omega_a^{\alpha\beta} \xi_\beta, \quad (4.87)$$

$$\delta \bar{\xi}^{\dot{\alpha}} = i \kappa [(\bar{\xi} \tilde{\Pi})^\gamma \Omega_\gamma^{\dot{\alpha}\beta} + \bar{c} i (\xi \Pi)_\gamma \Omega^{\gamma\dot{\alpha}\beta}] \bar{\xi}_\beta + \kappa [(\bar{\xi} \tilde{\sigma}^a \mathcal{D}) + \bar{c} i (\xi \sigma^a \tilde{\mathcal{D}})] \Omega_a^{\dot{\alpha}\beta} \bar{\xi}_\beta, \quad (4.88)$$

$$\delta \pi_\alpha = -i \kappa [(\bar{\xi} \tilde{\Pi})^\gamma \Omega_{\gamma\alpha\beta} + \bar{c} i (\xi \Pi)_\gamma \Omega^{\gamma\alpha\beta}] \pi^\beta - \kappa [(\bar{\xi} \tilde{\sigma}^a \mathcal{D}) + \bar{c} i (\xi \sigma^a \tilde{\mathcal{D}})] \Omega_{a\alpha\beta} \pi^\beta + \kappa \bar{c} i (\Pi \tilde{\mathcal{D}})_\alpha,$$

$$\delta \bar{\pi}_{\dot{\alpha}} = i \kappa [(\bar{\xi} \tilde{\Pi})^\gamma \Omega_{\gamma\dot{\alpha}\beta} + \bar{c} i (\xi \Pi)_\gamma \Omega^{\gamma\dot{\alpha}\beta}] \bar{\pi}^{\dot{\beta}} + \kappa [(\bar{\xi} \tilde{\sigma}^a \mathcal{D}) + \bar{c} i (\xi \sigma^a \tilde{\mathcal{D}})] \Omega_{a\dot{\alpha}\beta} \bar{\pi}^{\dot{\beta}} + \kappa (\mathcal{D} \Pi)_{\dot{\alpha}}.$$

Note that all terms on the right-hand sides of these transformations, except for the first terms on each line, contain constraints and are equal to zero in the weak sense. Also, one pays attention that the additional variables ξ^α , π_α and c.c. also transform under κ -transformations in the model under consideration.

5. Summary

In this paper, we have constructed a new superparticle model that describes the dynamics of a $4D$, $\mathcal{N}=1$ continuous spin superparticle in AdS_4 superspace.

The constructed model is a generalization of the continuous-spin superparticle model in flat $4D$, $\mathcal{N}=1$ superspace proposed in our paper [21]. We have started with considering a continuous spin superparticle in an arbitrary curved $4D$, $\mathcal{N}=1$ superspace. The analysis showed that in this case we should restrict the bosonic sector of the $4D$, $\mathcal{N}=1$ superspace to the constant curvature space. As a result of local supersymmetry, the continuous spin superparticle model turns out to be consistent only in the background superspace AdS_4 , whose supercurvature and supertorsion are determined by expressions (4.12). The Lagrangian and the bosonic and fermionic constraints of the model are constructed in explicit form.

After fixing the type of superspace, the Poisson brackets of the fermionic constraints $\mathcal{D}_\alpha \approx 0$, $\bar{\mathcal{D}}_{\dot{\alpha}} \approx 0$ (3.38) have the form (4.45). A straightforward calculation shows that the rank of the matrix of these Poisson brackets is three. Therefore, in contrast to the flat case, only one fermionic constraint (namely, $F_1 \approx 0$ (4.77)) of the four fermionic constraints is the first-class constraint, while the other three ($F_2 \approx 0$ (4.80) and $\mathcal{G}_1 \approx 0$, $\mathcal{G}_2 \approx 0$ (4.81)) are the second-class constraints. Let us pay attention to that it is precisely the presence of commuting spinor additional variables in the model under consideration that made it possible to carry out such a separation of four fermionic constraints in a covariant manner.

The presence of unequal numbers of first- and second-class fermionic constraints may cause some confusion. For example, in the well-known Brink-Schwarz superparticle model [2] the number of fermionic κ -symmetries is equal to half the number of fermionic covariant derivatives (or supersymmetry generators). The same is in the continuous spin superparticle model in flat superspace [21]. However, the supersymmetric models with a different number of fermionic κ -symmetries were also considered earlier in the literature, mainly in the context of BPS states (see, e.g., [55]). In particular, in [56,57], the superparticle models with three κ -symmetries were constructed for $4D$, $\mathcal{N}=1$ case (3/4 BPS states). The $4D$, $\mathcal{N}=1$ supersymmetric models with one κ -symmetry, which correspond to 1/4 BPS states, were studied in [58,59]. We also note the paper [60], where a superparticle model corresponding to the case of 1/4 BPS states and describing the model with $1D$, $\mathcal{N}=8$ supersymmetry was considered.

The model of the superparticle obtained here is very non-trivial for its quantization. In the flat case [21], the number of fermionic constraints was even and it was possible to separate them into two halves, complex conjugate to each other, and to carry out quantization according to the Gupta-Bleuler procedure. Now, an odd number of fermionic second-class constraints requires either the introduction of Dirac brackets for them (or for some of them) or the use of a procedure for converting second-class constraints into first-class ones by introducing additional (odd in our case) variables. Both of these methods require additional analysis, which is planned to be considered in subsequent works.

We expect that the specific structure of the fermionic constraints obtained here should lead, after quantization, to specific superfield equations of motion in AdS superspace. We assume that the form of such equations is related to specific representations of the $4D$, $\mathcal{N}=1$ anti-de Sitter $OSP(1|4)$ supergroup. This assumption certainly deserves study.

Another extremely interesting and promising direction for generalizing the model considered here is the construction of a model of the $4D$, $\mathcal{N}=2$ continuous spin superparticle. At present, almost all issues concerning such a model remain open. First, a description of continuous spin representations of the $4D$, $\mathcal{N}=2$ Poincaré supergroup in terms of the $\mathcal{N}=2$ superspace has not yet been given. Second, the procedure for the continuous spin superparticle coupling to the $\mathcal{N}=2$ AdS superspace is not entirely obvious.¹¹ We believe that

¹¹ The $4D$, $\mathcal{N}=2$ AdS superspace has been widely discussed in the literature (see, e.g., [19,20,61–63] and references therein).

the most natural way to construct the $4D$, $\mathcal{N}=2$ superparticle model can be realized in terms of harmonic superspace [64–67]. Related to the problem under consideration here, $\mathcal{N}=2$ higher spin field theories were studied in the recent papers [68–72]. Of particular relevance to our problems is a recent paper [73], which deals with superfield theories with $\mathcal{N}=2$ AdS supergroup $OSp(2|4)$ symmetry in harmonic superspace. These results, in principle can serve as a basis for constructing the $4D$, $\mathcal{N}=2$ continuous spin superparticle model in the AdS harmonic superspace.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Sergey Fedoruk reports a relationship with Joint Institute for Nuclear Research that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

We are grateful to Evgeny Ivanov for useful comments.

Appendix A. Vectors Π^a and $\mathcal{P}^a$ on the constraints surface

We introduce the vectors

$$n_+^a := \xi \sigma^a \bar{\xi}, \quad n_-^a := \pi \sigma^a \bar{\pi}, \quad m^a := \xi \sigma^a \bar{\pi}, \quad \bar{m}^a := \pi \sigma^a \bar{\xi}, \quad (\text{A.1})$$

that form a basis in the vector space. Their nonzero products have the form

$$(n_+ \cdot n_-) = -2N\bar{N}, \quad (m \cdot \bar{m}) = 2N\bar{N}, \quad (\text{A.2})$$

where we use the following conventions for the products of 4-vectors: $(n_\pm \cdot m) := n_\pm^a m_a$, etc.

The constraints (4.34) and (4.35) rewritten in the form (here we consider all relations on the constraint surface; therefore, we use the sign “=” instead of the sign “ $\approx$ ” of the weak equality)

$$(\Pi \cdot n_\pm) = \mu, \quad (\text{A.3})$$

where the constraint (4.37) has the form

$$(\Pi \cdot \Pi) = 0. \quad (\text{A.4})$$

with the vector Π_a defined in (4.33).

The vector Π_a (4.33) can be expanded on the space of the constraints in the basis vectors (A.1):

$$\Pi^a = A^+ n_+^a + A^- n_-^a + B m^a + \bar{B} \bar{m}^a, \quad (\text{A.5})$$

where $A^\pm$, B and $\bar{B}$ are some quantities. Let us define them.

The conditions (A.3) lead to $A^\pm = -\frac{\mu}{2N\bar{N}}$. Then, the condition (A.4) gives $B = \frac{\bar{c}\mu}{2N\bar{N}}$ and $\bar{B} = \frac{c\mu}{2N\bar{N}}$, where $c\bar{c} = 1$. Thus, on the constraint surface the following decomposition takes place:

$$\Pi^a = -\frac{\mu}{2N\bar{N}}(n_+^a + n_-^a) + \frac{\mu}{2N\bar{N}}(\bar{c}m^a + c\bar{m}^a). \quad (\text{A.6})$$

It is important that, as follows from (A.6), the equalities $(\Pi \cdot m) = c\mu$ and $(\Pi \cdot \bar{m}) = \bar{c}\mu$ are satisfied on the constraint surface. That is, the following constraints:

$$(\xi \Pi \bar{\pi}) = c\mu, \quad (\pi \Pi \bar{\xi}) = \bar{c}\mu \quad (\text{A.7})$$

take place.

Note that the constraint (4.31) of the form $\mathcal{U} = \xi^\alpha \pi_\alpha - \bar{\pi}_{\dot{\alpha}} \bar{\xi}^{\dot{\alpha}} \approx 0$ generates the transformations

$$\xi^\alpha \rightarrow e^{i\phi} \xi^\alpha, \quad \bar{\xi}^{\dot{\alpha}} \rightarrow e^{-i\phi} \bar{\xi}^{\dot{\alpha}}, \quad \pi_\alpha \rightarrow e^{-i\phi} \pi_\alpha, \quad \bar{\pi}_{\dot{\alpha}} \rightarrow e^{i\phi} \bar{\pi}_{\dot{\alpha}}, \quad (\text{A.8})$$

where $0 \leq \phi < 2\pi$ is the local angle parameter. As we can see from the expressions (A.7), the local $U(1)$ -symmetry (A.8) allows us to fix the constant c , for example, by taking $c = -i$.

Also note that after using (4.33) and (A.6), we get a useful decomposition of the covariant momentum in the form

$$\mathcal{P}^a = \frac{1}{2} \left(x - \frac{\mu}{2N\bar{N}} \right) (n_+^a + n_-^a) + \frac{\mu}{2N\bar{N}} (\bar{c}m^a + c\bar{m}^a). \quad (\text{A.9})$$

Appendix B. Poisson brackets of covariant momenta in AdS superspace

In this Appendix we present the Poisson brackets in AdS superspace for basic quantities.

Using the direct calculation and relations

$$\{M_{ab}, (\xi\sigma^c\bar{\xi})\} = \delta_a^c(\xi\sigma_b\bar{\xi}) - \delta_b^c(\xi\sigma_a\bar{\xi}),$$

we obtain that in AdS superspace the Poisson brackets (3.22) take the form:

$$\{P_a, P_b\} = \kappa^2 M_{ab} + 2\Omega_{[ab]}{}^c P_c, \quad (B.1)$$

$$\{\mathcal{D}_\alpha, \mathcal{D}_\beta\} = -4\kappa M_{\alpha\beta} + 2i\Omega_{(\alpha\beta)}{}^\gamma \mathcal{D}_\gamma, \quad (B.2)$$

$$\{\tilde{\mathcal{D}}_\alpha, \tilde{\mathcal{D}}_\beta\} = 4\kappa \tilde{M}_{\alpha\beta} - 2i\Omega_{(\alpha\beta)}{}^\gamma \tilde{\mathcal{D}}_\gamma, \quad (B.3)$$

$$\{\mathcal{D}_\alpha, \tilde{\mathcal{D}}_\beta\} = -2i(\sigma^c)_{\alpha\beta} P_c + i\Omega_{\beta\alpha}{}^\gamma \mathcal{D}_\gamma - i\Omega_{\alpha\beta}{}^\gamma \tilde{\mathcal{D}}_\gamma, \quad (B.4)$$

$$\{P_a, \mathcal{D}_\alpha\} = \frac{\kappa}{2}(\sigma_a)_{\alpha\dot{\gamma}} \tilde{\mathcal{D}}^{\dot{\gamma}} + \Omega_{a\alpha}{}^\beta \mathcal{D}_\beta - i\Omega_{a\alpha}{}^b P_b, \quad (B.5)$$

$$\{P_a, \tilde{\mathcal{D}}_\alpha\} = \frac{\kappa}{2}(\sigma_a)_{\gamma\dot{\alpha}} \mathcal{D}^\gamma - \Omega_{a\dot{\alpha}}{}^\beta \tilde{\mathcal{D}}_\beta - i\Omega_{a\dot{\alpha}}{}^b P_b. \quad (B.6)$$

For the quantity (4.33) the Poisson brackets are

$$\{\Pi_a, \Pi_b\} = 2\Omega_{[ab]}{}^c \Pi_c, \quad (B.7)$$

$$\{\Pi_a, \mathcal{D}_\alpha\} = \frac{\kappa}{2}(\sigma_a)_{\alpha\dot{\gamma}} \tilde{\mathcal{D}}^{\dot{\gamma}} + \Omega_{a\alpha}{}^\beta \mathcal{D}_\beta - i\Omega_{a\alpha}{}^b \Pi_b, \quad (B.8)$$

$$\{\Pi_a, \tilde{\mathcal{D}}_\alpha\} = \frac{\kappa}{2}(\sigma_a)_{\gamma\dot{\alpha}} \mathcal{D}^\gamma - \Omega_{a\dot{\alpha}}{}^\beta \tilde{\mathcal{D}}_\beta - i\Omega_{a\dot{\alpha}}{}^b \Pi_b. \quad (B.9)$$

Also, one can calculate the following Poisson brackets:

$$\{\xi^\beta, \mathcal{D}_\alpha\} = i\Omega_{\alpha\gamma}{}^\beta \xi^\gamma, \quad \{\xi^\beta, \tilde{\mathcal{D}}_\alpha\} = i\Omega_{\alpha\dot{\gamma}}{}^\beta \xi^\gamma, \quad (B.10)$$

$$\{\bar{\xi}^{\dot{\beta}}, \mathcal{D}_\alpha\} = i\Omega_{\alpha\dot{\gamma}}{}^{\dot{\beta}} \bar{\xi}^{\dot{\gamma}}, \quad \{\bar{\xi}^{\dot{\beta}}, \tilde{\mathcal{D}}_\alpha\} = i\Omega_{\alpha\gamma}{}^{\dot{\beta}} \bar{\xi}^{\dot{\gamma}}, \quad (B.11)$$

$$\{\xi^\beta, \Pi_a\} = \Omega_{a\gamma}{}^\beta \xi^\gamma - \frac{\kappa}{2}(\bar{\pi}\tilde{\sigma}_a)^\beta, \quad \{\bar{\xi}^{\dot{\beta}}, \Pi_a\} = \Omega_{a\dot{\gamma}}{}^{\dot{\beta}} \bar{\xi}^{\dot{\gamma}} - \frac{\kappa}{2}(\tilde{\sigma}_a\pi)^{\dot{\beta}}. \quad (B.12)$$

These relations are used in the analysis of the constraints in the model under consideration.

References

- [1] R. Casalbuoni, The classical mechanics for Bose-Fermi systems, *Nuovo Cim. A* 33 (1976) 389.
- [2] L. Brink, J.H. Schwarz, Quantum superspace, *Phys. Lett. B* 100 (1981) 310.
- [3] J.A.D. Azcarraga, J. Lukierski, Supersymmetric particles with internal symmetries and central charges, *Phys. Lett. B* 113 (1982) 170.
- [4] W. Siegel, Hidden local supersymmetry in the supersymmetric particle action, *Phys. Lett. B* 128 (1983) 397.
- [5] L. Lusanna, B. Milewski, $\mathcal{N}=2$ Super-Yang-Mills and supergravity constraints from coupling to a supersymmetric particle, *Nucl. Phys. B* 247 (1984) 396.
- [6] A. Frydryszak, $\mathcal{N}$ -extended free superfields ($N=2,4,6,8$) from quantization of a supersymmetric particle model, *Phys. Rev. D* 30 (1984) 2172.
- [7] J.A.D. Azcarraga, J. Lukierski, Supersymmetric particle model with additional bosonic coordinates, *Z. Phys. C* 30 (1986) 221.
- [8] E. Witten, Twistor-like transform in ten-dimensions, *Nucl. Phys. B* 266 (1986) 245.
- [9] J.A. Shapiro, C.C. Taylor, Supergravity torsion constraints from the 10D superparticle, *Phys. Lett. B* 181 (1986a) 67.
- [10] J.A. Shapiro, C.C. Taylor, Superspace supergravity from the superstring, *Phys. Lett. B* 181 (1986b) 69.
- [11] L. Brink, M. Henneaux, C. Teitelboim, Covariant Hamiltonian formulation of the superparticle, *Nucl. Phys. B* 293 (1987) 505.
- [12] W. Siegel, The superparticle revised, *Phys. Lett. B* 203 (1988) 79.
- [13] J.A. Shapiro, C.C. Taylor, The space-time supersymmetric formulation of the string, *Phys. Rept.* 191 (1990) 221.
- [14] I.A. Bandos, J. Lukierski, C. Preitschopf, D.P. Sorokin, Osp supergroup manifolds, superparticles and supertwistors, *Phys. Rev. D* 61 (2000) 65009. [arXiv:hep-th/9907113](#)
- [15] I.A. Bandos, E. Ivanov, J. Lukierski, D.P. Sorokin, On the superconformal flatness of AdS superspaces, *JHEP* 06 (2002) p. 40. [arXiv:hep-th/0205104](#)
- [16] I.A. Bandos, J.A.D. Azcarraga, J.M. Izquierdo, J. Lukierski, $D=4$ Supergravity dynamically coupled to a massless superparticle in a superfield Lagrangian approach, *Phys. Rev. D* 67 (2003) 65003. [arXiv:hep-th/0207139](#)
- [17] S. Fedoruk, E. Ivanov, J. Lukierski, Massless higher spin $D=4$ superparticle with both $N=1$ supersymmetry and its bosonic counterpart, *Phys. Lett. B* 641 (2006) 226. [arXiv:hep-th/0606053](#)
- [18] I.L. Buchbinder, I.B. Samsonov, $\mathcal{N}=3$ Superparticle model, *Nucl. Phys. B* 802 (2008) 108. [Hep-th, arXiv:0801.4907](#)
- [19] N.E. Koning, S.M. Kuzenko, E.S.N. Raptakis, The anti-de Sitter supergeometry revised, *JHEP* 02 (2025) p. 175. [arXiv:2412.03172](#)
- [20] N.E. Koning, E.S.N. Raptakis, New Superparticle Models in AdS Superspaces, 2025 [arXiv:2506.17897](#).
- [21] I.L. Buchbinder, S.A. Fedoruk, Continuous Spin Superparticle Model, 2025 [arXiv:2506.19709](#).
- [22] I.L. Buchbinder, S. Fedoruk, A.P. Isaev, A. Rusnak, Model of massless relativistic particle with continuous spin and its twistorial description, *JHEP* 07 (2018) 31. [Hep-th, arXiv:1805.09706](#)
- [23] I.L. Buchbinder, S.A. Fedoruk, A.P. Isaev, V.A. Krykhtin, Infinite (continuous) spin particle in constant curvature space, *Phys. Lett. B* 853 (2024) 138689. [Hep-th, arXiv:2402.13879](#)
- [24] I.L. Buchbinder, S.A. Fedoruk, A.P. Isaev, M.A. Podoinitsyn, On the realization of the infinite (continuous) spin field representations in AdS_4 space, *Phys. Lett. B* 861 (2025) 139226. [Hep-th, arXiv:2410.07873](#)
- [25] R.R. Metsaev, BRST-BV approach to continuous-spin field, *Phys. Lett. B* 781 (2018) 568–573. [Hep-th, arXiv:1803.08421](#)
- [26] R.R. Metsaev, Light-cone gauge massive and partially-massless fields in $AdS(4)$, *Phys. Lett. B* 839 (2023) 137790. [Hep-th, arXiv:2212.14728](#)
- [27] R.R. Metsaev, Interacting Massive/Massless Continuous-Spin Fields and Integer-Spin Fields, 2025a. [arXiv:2505.02817](#).
- [28] R.R. Metsaev, Light-Cone Vector Superspace and Continuous-Spin Field in AdS, 2025b. [arXiv:2507.05194](#).
- [29] I.L. Buchbinder, V.A. Krykhtin, H. Takata, BRST approach to Lagrangian construction for bosonic continuous spin field, *Phys. Lett. B* 785 (2018) 315. [Hep-th, arXiv:1806.01640](#)

- [30] I.L. Buchbinder, S.A. Fedoruk, A.P. Isaev, V.A. Krykhtin, BRST construction for infinite spin field on AdS_4 , Eur. Phys. J. Plus 139 (2024) 621. Hep-th, [arXiv:2403.14446](#)
- [31] P. Schuster, N. Toro, K. Zhou, Interactions of particles with “continuous spin” fields, JHEP 04 (2023) p. 10. [arXiv:2303.04816](#)
- [32] P. Schuster, N. Toro, Quantum electrodynamics mediated by a photon with continuous spin, Phys. Rev. D 109 (2024) 96008. Hep-th, [arXiv:2308.16218](#)
- [33] P. Schuster, G. Sundaresan, N. Toro, Thermodynamics of continuous spin photons, Phys. Rev. D 111 (2025) 56019. Hep-ph, [arXiv:2406.14616](#)
- [34] S. Kundu, P. Schuster, N. Toro, A First Look at “Continuous Spin” Gravity-Time Delay Signature, 2025a. [arXiv:2503.03817](#).
- [35] S. Kundu, A. Russo, P. Schuster, N. Toro, Interactions of a Continuous-Spin Field with Spin-1/2 Particle, 2025b. [arXiv:2505.14770](#).
- [36] I.L. Buchbinder, S.J. Gates, K. Koutrolkos, Superfield continuous spin equations of motion, Phys. Lett. B 793 (2019a) 445. Hep-th, [arXiv:1903.08631](#)
- [37] I.L. Buchbinder, M.V. Khabarov, T.V. Snegirev, Y.M. Zinoviev, Lagrangian formulation for the infinite spin $N = 1$ supermultiplets in $d = 4$, Nucl. Phys. B 946 (2019b) 114717. Hep-th, [arXiv:1904.05580](#)
- [38] R.R. Metsaev, Cubic interactions for arbitrary spin N -extended massless supermultiplets in 4D flat space, JHEP 11 (2019) 84. Hep-th, [arXiv:1909.05241](#)
- [39] M. Najafizadeh, Supersymmetric continuous spin gauge theory, JHEP 03 (2020) p. 27. [arXiv:1912.12310](#)
- [40] M. Najafizadeh, Off-shell supersymmetric continuous spin gauge theory, JHEP 02 (2022) p. 38. [arXiv:2112.10178](#)
- [41] I.L. Buchbinder, S. Fedoruk, A.P. Isaev, V.A. Krykhtin, On the off-shell superfield Lagrangian formulation of 4D, $N=1$ supersymmetric infinite spin theory, Phys. Lett. B 829 (2022) 137139. Hep-th, [arXiv:2203.12904](#)
- [42] J. Wess, J. Bagger, *Supersymmetry and Supergravity*, Princeton University Press, 1983.
- [43] I.L. Buchbinder, S.M. Kuzenko, *Ideas and Methods of Supersymmetry and Supergravity, or a Walk Through Superspace*, IOP, Bristol, Revised Edition 1998, 1995.
- [44] E.A. Ivanov, A.S. Sorin, Superfield formulation of $OSp(1,4)$ supersymmetry, J. Phys. A Math. Gen. 13 (1980) 1159.
- [45] B.W. Keck, An alternative class of supersymmetries, J. Phys. A 8 (1975) 1819.
- [46] B. Zumino, Nonlinear realization of supersymmetry in de Sitter space, Nucl. Phys. B 127 (1977) 189.
- [47] E.A. Ivanov, A.S. Sorin, Supergroup $OSp(1,4)$ and classical solutions of the Wess-Zumino model, Theor. Math. Phys. 39 (1979a) 394.
- [48] E.A. Ivanov, A.S. Sorin, Wess-Zumino model as linear sigma model of spontaneously broken conformal and $Osp(1,4)$ supersymmetries, Sov. J. Nucl. Phys. 30 (1979b) 440.
- [49] D.M. Gitman, I.V. Tyutin, *Quantization of Fields with Constraints*, Springer, 1990.
- [50] K. Pilch, P.V. Nieuwenhuizen, M.F. Sohnius, De Sitter superalgebras and supergravity, Commun. Math. Phys. 98 (1985) 105.
- [51] T. Anous, D.Z. Freedman, A. Maloney, De Sitter supersymmetry revised, JHEP 07 (2014) 119. Hep-th, [arXiv:1403.5038](#)
- [52] E.A. Bergshoeff, D.Z. Freedman, R. Kallosh, A.V. Proeyen, Pure de Sitter supergravity, Phys. Rev. D 92 (2015) 85040. Hep-th, [arXiv:1507.08264](#)
- [53] S.M. Kuzenko, Complex linear Goldstino superfield and supergravity, JHEP 10 (2015) 6. Hep-th, [arXiv:1508.03190](#)
- [54] I.L. Buchbinder, S.A. Fedoruk, A.P. Isaev, V.A. Krykhtin, On BRST Lagrangian formulation of massless higher spin fields, Russ. Phys. J. 67 (2024) 1806. Hep-th, [arXiv:2412.08298](#)
- [55] J.P. Gauntlett, G.W. Gibbons, C.M. Hull, P.K. Townsend, BPS states of $D=4$, $\mathcal{N}=1$ supersymmetry, Commun. Math. Phys. 216 (2001) 431. [arXiv:hep-th/0001024](#)
- [56] I.A. Bandos, J. Lukierski, Tensorial central charges and new superparticle models with fundamental spinor coordinates, Mod. Phys. Lett. A 14 (1999) 1257–1272. [arXiv:hep-th/9811022](#)
- [57] I.A. Bandos, J. Lukierski, D.P. Sorokin, Superparticle models with tensorial central charges, Phys. Rev. D 61 (2000) 45002. [arXiv:hep-th/9904109](#)
- [58] F. Delduc, E. Ivanov, S. Krivonos, 1/4 Partial breaking of global supersymmetry and new superparticle actions, Nucl. Phys. B 576 (2000) 196. [arXiv:hep-th/9912222](#)
- [59] S. Fedoruk, V.G. Zima, Massive superparticle with tensorial central charges, Mod. Phys. Lett. A 15 (2000) 2281. [arXiv:hep-th/0009166](#)
- [60] S. Bellucci, A. Galajinsky, E. Ivanov, S. Krivonos, Quantum mechanics of superparticle with 1/4 supersymmetry breaking, Phys. Rev. D 65 (2002) 104023. [arXiv:hep-th/0112075](#)
- [61] S.M. Kuzenko, G. Tartaglino-Mazzucchelli, Supertwistor realisations of AdS superspaces, Eur. Phys. J. C 82 (2) (2022) 146. Hep-th, [arXiv:2108.03907](#)
- [62] N.E. Koning, S.M. Kuzenko, E.S.N. Raptakis, Embedding formalism for $\mathcal{N}$ -extended AdS superspace in four dimensions, JHEP 11 (2023) p. 63. [arXiv:2308.04135](#)
- [63] S.M. Kuzenko, K. Turner, Embedding formalism for (p,q) AdS superspaces in three dimensions, JHEP 06 (2023) p. 142. [arXiv:2303.03082](#)
- [64] A. Galperin, E. Ivanov, S. Kalitzin, V. Ogievetsky, E. Sokatchev, Unconstrained $\mathcal{N}=2$ Matter, Yang-Mills and supergravity theories in harmonic superspace, Class. Quant. Grav. 1 (1984) 469. [erratum: Class. Quant. Grav. 2 (1985) 127]
- [65] A.S. Galperin, N.A. Ky, E. Sokatchev, $\mathcal{N}=2$ Supergravity in superspace: solution to the constraints, Class. Quant. Grav. 4 (1987a) 1235.
- [66] A.S. Galperin, E.A. Ivanov, V.I. Ogievetsky, E. Sokatchev, $\mathcal{N}=2$ Supergravity in superspace: different versions and matter couplings, Class. Quant. Grav. 4 (1987) 1255.
- [67] A.S. Galperin, E.A. Ivanov, V.I. Ogievetsky, E. Sokatchev, *Harmonic Superspace*, Cambridge Monographs on Mathematical Physics, Cambridge University Press, 2001.
- [68] I. Buchbinder, E. Ivanov, N. Zaigraev, Unconstrained off-shell superfield formulation of 4D, $\mathcal{N}=2$ supersymmetric higher spins, JHEP 12 (2021) 16. Hep-th, [arXiv:2109.07639](#)
- [69] I. Buchbinder, E. Ivanov, N. Zaigraev, Off-shell cubic hypermultiplet couplings to $\mathcal{N}=2$ higher spin gauge superfields, JHEP 104 (05) (2022). Hep-th, [arXiv:2202.08196](#).
- [70] I. Buchbinder, E. Ivanov, N. Zaigraev, $\mathcal{N}=2$ Higher spins: superfield equations of motion, the hypermultiplet supercurrents, and the component structure, JHEP 03 (2023) p. 36. [arXiv:2212.14114](#)
- [71] I. Buchbinder, E. Ivanov, N. Zaigraev, $\mathcal{N}=2$ Superconformal higher-spin multiplets and their hypermultiplet couplings, JHEP 08 (2024) 120. Hep-th, [arXiv:2404.19016](#)
- [72] I. Buchbinder, E. Ivanov, N. Zaigraev, Towards $\mathcal{N}=2$ Higher-Spin Supergravity, 2025. [arXiv:2503.02438](#).
- [73] E. Ivanov, N. Zaigraev, $\mathcal{N}=2$ AdS Hypermultiplets in Harmonic Superspace, 2025. [arXiv:2509.01406](#).